

Review

A Review of Crop Attribute Detection for Agricultural Harvesting Machinery

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Abstract

Crop attribute detection, as a key component of intelligent agricultural harvesting machinery, plays a crucial role in harvesting efficiency, loss reduction, and autonomous operation control. Compared with existing reviews on artificial intelligence and sensing technologies in agriculture, this review focuses on crop attribute detection scenarios oriented toward the intelligent decision-making and control requirements of agricultural harvesting machinery. It mainly analyzes crop attributes that affect harvesting operations, as well as the sensors and algorithms involved in detecting these attributes, and further clarifies the relationship between detection methods and control decisions in agricultural harvesting machinery. For grain crops, the key attributes relevant to harvesting operations include plant height, plant density, spike number, crop lodging, canopy structure, and crop position. For fruit and vegetable crops, the key attributes relevant to harvesting operations include maturity, position, and quality. From the perspectives of multi-source data acquisition, data analysis, and attribute detection algorithms, the key technologies in the field of crop attribute detection are systematically summarized and analyzed, including sensors used in crop attribute detection, such as RGB, spectral, near-infrared, and LiDAR sensors, as well as data analysis and recognition approaches, such as image classification, object detection, and point cloud analysis. The complexity of field environments and the dynamics of machine operation are analyzed, highlighting the technical bottlenecks of current detection systems in environmental adaptability, real-time responsiveness, and resistance to interference. To address these challenges, feasible optimization directions were proposed, including multi-sensor fusion, weakly supervised learning, and few-shot learning. This review aims to provide systematic references and theoretical support for the coordinated development of crop detection and control decision-making in intelligent agricultural harvesting systems.

Keywords: crop attribute detection; intelligent harvesting; multi-sensor fusion; machine vision

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1. Introduction

Crops, as the fundamental units of agricultural production, directly affect food security and economic development through their growth status and yield. Agricultural

harvesting machinery performs harvesting, threshing, and sorting, and is widely applied to wheat, rice, corn, and fruit and vegetable crops [1]. Traditional machinery relies heavily on manual operation and static parameters, with poor adaptability to complex harvesting environments, leading to low efficiency, high losses, and resource waste [2]. With the advancement of intelligent agriculture, agricultural harvesting machinery is gaining enhanced capabilities in autonomous perception, intelligent decision-making, and dynamic execution. The enhanced capabilities serve as key factors in boosting productivity and quality control. Therefore, developing agricultural harvesting machinery with high-precision perception and intelligent operation has become an important direction for technological innovation in agricultural equipment.

During harvesting operations, crop attribute detection plays a crucial role. Harvesting machinery can dynamically adjust operational parameters based on perception results, thereby enabling adaptive control of the operation process. The adjustment optimizes operation paths, reduces harvest loss rates, and improves the marketable rate of crops [3]. In practical intelligent agricultural machinery systems, the effective utilization of detection information depends not only on perception accuracy but also on efficient data exchange and system integration mechanisms. Standardized communication protocols (such as ISOBUS, ISO 11783) provide a unified framework for information interaction between perception modules and the control bus of tractors or harvesters [4]. Through such protocols, detection results can be transmitted to onboard control units in real time, thereby enabling coordinated operation among perception, decision-making, and execution. Existing studies have demonstrated that integrating perception systems with agricultural machinery communication architectures is of great significance for achieving intelligent control and automated operations.

Existing reviews on artificial intelligence and sensing technologies in agriculture have mainly focused on general agricultural scenarios, such as crop monitoring, disease identification, yield prediction, farm management, crop phenotyping, or specific sensing and detection methods. These reviews mainly emphasize detection methods themselves, with relatively limited connections to practical operational scenarios and control requirements [1]. For example, Liakos et al. reviewed machine learning applications in agriculture, covering tasks such as yield prediction, disease and weed detection, crop quality assessment, and water management [2]. However, throughout the agricultural production process, including cultivation, management, and harvesting, the same detection information may serve different requirements and objectives under different operational scenarios. For the specific field of crop attribute detection in agricultural harvesting machinery, there is still a lack of review studies, especially those summarizing crop attribute detection methods in multiple harvesting scenarios such as fields and greenhouses from the perspectives of harvesting efficiency, loss rate, operating speed, and failure rate. Such studies are important for promoting intelligent, unmanned, and precise harvesting operations. Therefore, this review takes crop attribute detection in agricultural harvesting environments as its focus and summarizes its role in supporting perception, decision-making, and control for intelligent harvesting machinery. It first analyzes the factors affecting harvesting operations in multiple agricultural harvesting scenarios, including field and greenhouse environments. On this basis, it summarizes multiple attributes of grain, fruit, and vegetable crops that affect harvesting operations, including plant height, plant density, spike number, lodging, canopy structure, crop position, maturity, position, and quality. By considering the influence of crop attributes on the accuracy and efficiency of control decisions, this review further summarizes and analyzes methods such as image segmentation, image classification, object detection, and point cloud analysis. It also explains how detection methods and results can support practical machinery operations, such as header

height adjustment, forward speed regulation, feed rate control, target localization, picking path planning, and end-effector control.

Research on crop detection in agricultural harvesting has developed multiple technical routes, mainly including visible-light image-based feature analysis, multi-source sensor perception, and data-driven intelligent recognition methods[5]. Early methods based on RGB imaging and handcrafted features were easily affected by changes in environmental conditions[6]. The application of sensors such as LiDAR, near-infrared, and hyperspectral imaging has enabled the perception of crop three-dimensional structures and physiological characteristics[7]. Combined with deep learning methods such as object detection (e.g., YOLO) and instance segmentation (e.g., Mask R-CNN), detection accuracy and robustness have been significantly improved, and detection tasks have expanded to include spike counting, lodging detection, disease localization, and quality grading[8]. For example, a Mask R-CNN-based system achieved 77.76% spike recognition, 98.81% disease detection accuracy, and 77.19% disease severity prediction accuracy in wheat[9]. These detection results can provide quantitative information for harvesting control, such as adjusting feed rate based on crop density and spike distribution estimation, conducting header height adjustment and path correction according to lodging-area identification, and performing picking-trajectory planning and end-effector control based on fruit target localization. In addition, detection accuracy and inference speed jointly determine whether perception results can be converted into control actions in time under continuous harvesting or robotic picking rhythms. To address challenges such as complex field conditions, real-time response requirements, and limited onboard computing resources, more researchers have begun to introduce methods such as multi-source data fusion, weakly supervised learning, and edge computing. These methods improve the robustness of crop attribute detection and the feasibility of field deployment through multimodal information complementarity, reduced annotation costs, and edge-side real-time inference[10].

Although current crop attribute detection technologies for agricultural harvesting scenarios have achieved promising results in some applications, their use for practical harvesting decision-making and control still faces multiple challenges, including complex weather conditions, variable illumination, dust, vibration, occlusion, and real-time response requirements, as summarized below: (1) factors such as variable field lighting, plant occlusion, and terrain undulation can easily introduce data noise and reduce detection accuracy[11]. Beyond general environmental complexity, extreme field conditions (such as rain, nighttime operation, and dusty environments) can further degrade sensor data quality, thereby significantly affecting the stability of crop attribute detection. (2) mechanical vibrations and posture changes during high-speed operations interfere with sensor data acquisition, causing errors in recognition and positioning. In addition, strict real-time requirements in harvesting limit the application of high-precision methods due to inference delays[12]. In response to these challenges, more researchers have begun to focus on adaptive multi-sensor fusion, lightweight models, and motion compensation, which may provide potential solutions for improving the robustness, adaptability, and real-time performance of detection systems in the future.

Based on the above background, this paper conducts a systematic review of key issues in crop attribute detection during agricultural harvesting operations. Unlike existing reviews that mainly focus on general agricultural monitoring or single detection tasks, this study constructs an application-oriented analytical framework tailored to harvesting scenarios by linking crop attribute requirements, sensing technologies, and detection methods under the dynamic and real-time constraints of harvesting operations, highlighting the relationship between perception and agricultural machinery control. Within this framework, crop attributes are regarded as perception targets, sensors provide the basis for data acquisition, and detection methods serve as tools for information interpretation.

The detected crop attributes act as information sources for harvesting decisions. For example, in field grain harvesting machinery, the detected attributes can be used for header height adjustment, forward speed regulation, and feed rate control; in fruit and vegetable harvesting scenarios such as greenhouses, orchards, or trellised cultivation systems, the detected attributes can provide a basis for target localization, picking path planning, and end-effector control. Meanwhile, practical deployment is also constrained by factors such as illumination variation, occlusion, dust, vibration, response speed, sensor cost, and system integration difficulty. Therefore, this review focuses on the relationship among crop attributes, sensors, and detection methods, and further relates them to harvesting decisions and deployment constraints to clarify the application value of crop attribute detection in intelligent harvesting operations. In addition, this study combines crop-type-based attribute classification with technology-pathway-based method classification to clarify how different detection methods serve specific operational objectives. Figure 1 presents a conceptual framework of crop attribute detection for intelligent agricultural harvesting, illustrating the relationships among sensing technologies, detection methods, crop attributes, and their roles in harvesting decision-making and machinery operation. Specifically, the objectives of this study include: (1) summarizing the key attribute requirements of grain crops and fruit and vegetable crops from the perspective of crop types, including plant height, canopy structure, density, spike positioning, and quality-related attributes; (2) reviewing the main sensing technologies and data analysis methods used in crop attribute detection from a technical perspective, including RGB imaging, LiDAR, multispectral, and near-infrared sensors, as well as image classification, object detection, image segmentation, and point cloud analysis methods in harvesting scenarios (Table 1); (3) summarizing the main challenges faced by current intelligent harvesting systems in terms of environmental adaptability, response speed, and interference resistance, and discussing future development directions such as sensor fusion, adaptive mechanisms, and big data integration, thereby providing systematic references for optimizing crop attribute detection technologies and intelligent harvesting equipment.

Table 1. Summary of typical methods.

Detection Method	Data Sensor	Specific Method	Applicable Crop	Detection Attribute	Performance and Practical Applicability	Reference
Image Segmentation	RGB Camera	Employ U-Net, Mask R-CNN for pixel-level segmentation of disease and lodging areas, handling occlusions and improving robustness via multi-view fusion.	Corn, Wheat, Fruits	Spike Area, Disease Spots, Lodging Area	Segmentation accuracy generally ranges from 80% to 95%. However, real-time performance is strongly affected by network size and image resolution, and robustness may decrease under blurred boundaries, occlusion, and uneven illumination.	[13–19]
Image Classification	RGB Camera, Near-Infrared Camera, Multispectral Camera	Use CNN, MobileNet, EfficientNet for end-to-end crop image classification by extracting texture and color features to identify varieties and maturity.	Fruits, Grains	Variety Identification, Maturity, Grade	MobileNet and EfficientNet typically achieve 80–95% accuracy. Lightweight models generally have lower inference latency and deployment cost, but their robustness may be limited when	[20–22]

Object Detection	RGB Camera, RGB-D Camera	Apply YOLOv5/v8, Faster R-CNN for precise detection and counting of crop targets (spikes, fruits, disease spots) under complex and fast-moving conditions.	Tomato, Corn	Fruit Location, Spike number, Disease Spots	maturity or quality differences are subtle under variable illumination.	[23–32]
					YOLO-based and Faster R-CNN models typically achieve detection accuracy of 75–90%, with R^2 values of 0.75–0.90 and RMSE values of 0.5–1.0. YOLO-type models generally have faster inference speed, making them more suitable for real-time target localization and counting.	
Point Cloud Analysis	LiDAR (Light Detection and Ranging), RGB-D Camera	Utilize LiDAR or RGB-D point clouds with stem fitting, point density, and CHM models to estimate plant height, lodging angle, and canopy structure.	Grains	Plant Height, Canopy, Lodging Status, Position	Plant height prediction typically achieves R^2 values of 0.80–0.95; lodging direction error generally ranges from 5° to 15°, and lodging severity error ranges from 5 to 15 cm. This method can support three-dimensional control tasks, but it involves relatively high sensor cost and data-processing load.	[33–35]
Other Methods	RGB Camera, LiDAR (Light Detection and Ranging), Multi-spectral Camera	Include methods like density regression (CSRNet), anomaly detection (AE/GAN), and physiological modeling for occlusion handling, boundary detection, and quality assessment.	Rice, Wheat, Fruits and Vegetables	Tillering Number, Quality Variation, Deformed Fruit Detection	Tillering number prediction typically achieves R^2 values of 0.60–0.80; density estimation error rates are generally within 5–15%, and anomaly detection accuracy ranges from 75% to 90%. Their real-time performance and robustness are strongly affected by task type, training data quality, and system integration complexity.	[36–37]

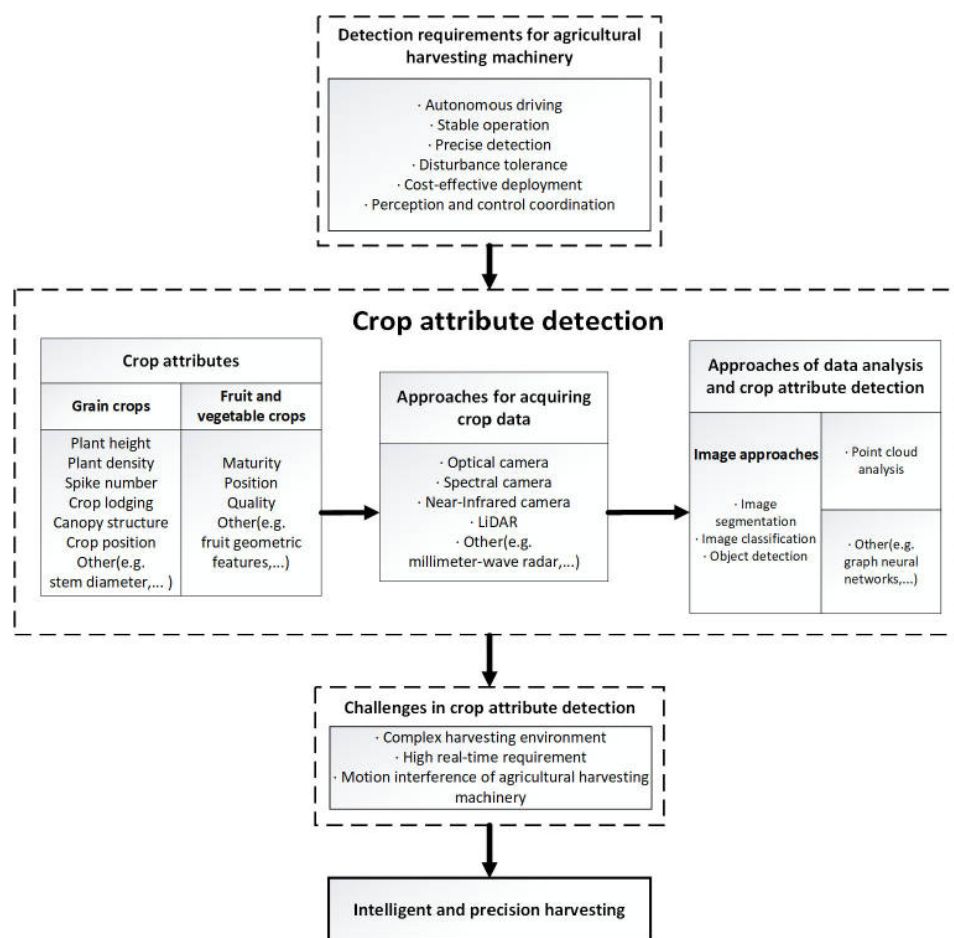


Figure 1. Conceptual framework of crop attribute detection for intelligent agricultural harvesting.

2. Methodology

This review adopted a structured literature review framework to improve the comprehensiveness and transparency of the collected studies. The literature search mainly covered relevant studies published between 2015 and 2025, and the final retrieval was completed in April 2026. This time range was selected to cover recent advances in agricultural harvesting machinery, crop attribute detection, sensor applications, and intelligent perception methods, while ensuring the timeliness and representativeness of the reviewed literature. Relevant publications were retrieved from multiple mainstream academic databases, including Web of Science, Scopus, IEEE Xplore, ScienceDirect, SpringerLink, and Google Scholar, supplemented by forward and backward tracing of key references. During retrieval, Web of Science and Scopus were mainly used to identify peer-reviewed journal articles and conference papers, IEEE Xplore was used to obtain engineering and technical studies, ScienceDirect and SpringerLink were used to retrieve full-text journal articles, and Google Scholar and reference tracing were used to identify additional relevant studies. Because some publications were indexed in more than one database, duplicate records were removed during screening. Both Chinese and English publications were considered, with a focus on representative studies related to agricultural harvesting scenarios, crop attribute detection tasks, sensing technologies, and practical application value.

During the retrieval process, multiple keyword combinations have been used, including “crop attribute detection”, “agricultural harvesting”, “harvesting machinery”, “machine vision”, “LiDAR”, “multispectral imaging”, “near-infrared”, “hyperspectral imaging”, “object detection”, “image segmentation”, “crop quality detection”, “crop monitoring”, “plant phenotyping”, “yield estimation”, “precision agriculture”, “smart

agriculture", "deep learning in agriculture", "edge computing in agriculture", "real-time detection", "canopy structure estimation", "crop height measurement", "plant density estimation", "fruit detection", "harvesting automation" and "agricultural robotics".

Based on the above keywords, Boolean search strings were further constructed to clarify the combination relationships among different keywords. The search strings mainly combined terms related to crop attribute detection, terms related to sensors and detection methods, and terms related to agricultural harvesting machinery application scenarios. Representative search strings included: (1) ("crop attribute detection" OR "crop trait detection" OR "plant phenotyping") AND ("agricultural harvesting" OR "harvesting machinery" OR "intelligent harvesting"); (2) ("machine vision" OR "RGB imaging" OR "LiDAR" OR "multispectral imaging" OR "hyperspectral imaging" OR "near-infrared spectroscopy") AND ("crop detection" OR "crop monitoring" OR "crop quality detection") AND ("harvesting" OR "harvesting machinery" OR "agricultural machinery"); (3) ("object detection" OR "image segmentation" OR "image classification" OR "point cloud analysis") AND ("fruit detection" OR "spike detection" OR "lodging detection" OR "plant height estimation" OR "plant density estimation") AND ("harvesting" OR "harvesting robot" OR "combine harvester"); and (4) ("deep learning" OR "multi-sensor fusion" OR "edge computing" OR "real-time detection") AND ("agricultural robotics" OR "harvesting automation" OR "smart agriculture"). The field settings, Boolean operators, and keyword combinations were appropriately adjusted according to the search rules of different databases.

The inclusion criteria were defined as follows: (1) studies related to crop attribute detection in agricultural harvesting scenarios; (2) studies involving sensing technologies, image analysis, spectral analysis, or point cloud analysis; (3) studies reporting crop attribute detection methods, detection results, or application value; and (4) studies providing information relevant to harvesting decision-making, machinery operation, or practical deployment. The exclusion criteria included: (1) studies unrelated to agricultural harvesting scenarios; (2) duplicate publications; (3) studies with insufficient methodological descriptions or unclear data sources; and (4) studies focusing only on general crop monitoring without a clear connection to harvesting operations.

The literature screening process was further organized into four steps. First, duplicate records from different databases were removed. Second, titles, abstracts, and keywords were screened to exclude studies unrelated to agricultural harvesting, crop attribute detection, sensing technologies, or intelligent perception. Third, potentially relevant studies were assessed through full-text reading, with emphasis on whether the research objects, detected attributes, sensor types, detection methods, evaluation results, and application scenarios were consistent with the scope of this review. Fourth, studies closely related to crop attributes, sensor technologies, detection methods, or machinery decision support in harvesting scenarios were retained for subsequent synthesis.

After screening and synthesis, approximately 300 relevant publications were thoroughly analyzed, and 147 representative studies were selected as the main references for this review. The distribution of the approximately 300 analyzed publications by year of publication is shown in Figure 2. The retained studies were mainly evaluated according to the following criteria: (1) relevance to agricultural harvesting scenarios; (2) clarity of the detected crop attributes and application tasks; (3) completeness of the description of sensors, algorithms, or data analysis methods; (4) availability of quantitative or qualitative evaluation results; (5) connection between the detected attributes and harvesting decisions or machinery operations; and (6) contribution to the classification framework of this review. The selected studies were further classified according to crop type, detected attribute, sensing technology, detection method, and application task, providing the basis for the subsequent systematic analysis and discussion.

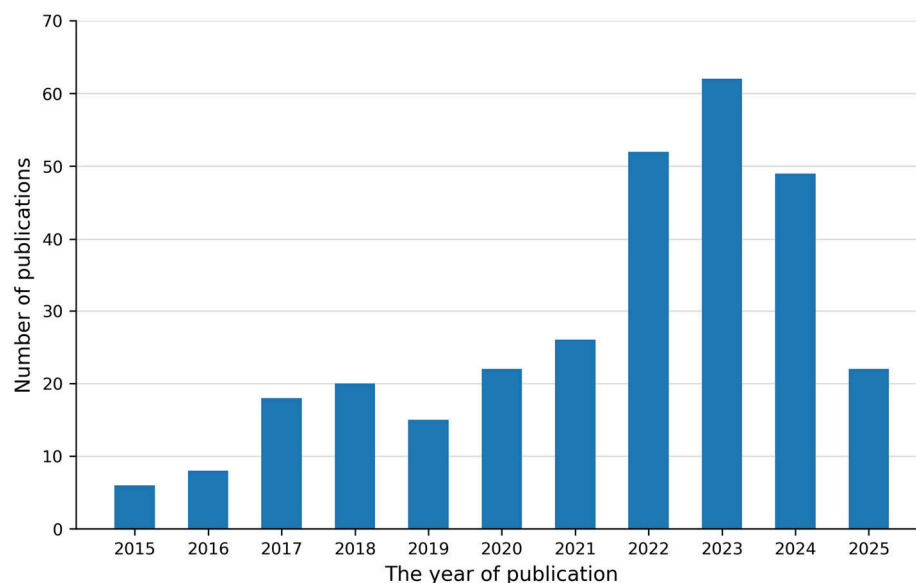


Figure 2. Distribution of analyzed publications by year of publication (2015–2025).

3. Crop Attributes for Agricultural Harvesting Machinery

In crop harvesting, attribute detection directly affects operational strategies and operational outcomes. The key factors influencing harvesting operations differ across scenarios. For field grain crops, factors such as crop density, lodging status, canopy distribution, and row structure can affect header height, forward speed, feed rate, path alignment, and machine load. For fruit and vegetable crops, factors such as maturity status, fruit position, external quality, canopy occlusion, and branch distribution can influence harvest timing, target selection, picking trajectory, end-effector motion, and grading decisions. Therefore, in this review, crop attributes are not discussed as independent biological indicators, but are regarded as perception variables directly corresponding to specific harvesting decisions. Based on these harvesting operation requirements, this chapter focuses on attributes of grain crops, including plant height, lodging status, density, spike number, canopy structure, and crop position, as well as attributes of fruit and vegetable crops, including maturity, spatial position, and quality, and analyzes their application value in improving harvesting efficiency, reducing losses, maintaining operational stability, and supporting machinery decision-making.

3.1. Attributes of Grain Crops

Grain crops such as rice, wheat, rapeseed, and maize are usually continuously and densely distributed in the field, and their attribute detection differs from perception tasks targeting individual objects in fruit and vegetable harvesting. Under conditions such as high-density planting and lodging, real-time perception of attributes such as plant height, density, spike number, lodging status, canopy structure, and spatial position is important for the control of agricultural harvesting machinery. This section focuses on the main grain crop attributes related to harvesting operations and their corresponding detection methods.

3.1.1. Plant Height

Plant height is a basic morphological indicator describing the vertical dimension of crop growth. In accordance with relevant national standards, it is measured from the base of the main stem at ground level to the upper end of the spike, with awns excluded from the measurement. This parameter is commonly used to characterize crop development,

biomass accumulation, and population structure[38]. Plant height varies with cultivar characteristics, growth stage, planting density, and environmental conditions, resulting in noticeable differences among crop species and growth periods.

In combine harvesting, crop height information is closely related to the setting of working components, especially the vertical position of the header. If the header is raised beyond the effective cutting range, part of the spike layer may remain unharvested, resulting in direct grain loss. Conversely, an overly low cutting position increases the amount of straw entering the threshing and cleaning units, which may raise the machine load and reduce threshing performance[39]. Therefore, continuous height sensing in fields with non-uniform growth or lodging conditions can provide a basis for coordinated regulation of forward speed, drum rotation speed, and feeding amount. Such parameter adjustment helps maintain a stable material flow through the harvester and reduces losses and grain damage during operation. In practical combine harvesting operations, plant height information needs to be updated fast enough to follow local changes in crop height along the travel path, so that the header can be adjusted before the cutter reaches the corresponding crop region. When height estimation is unstable under uneven terrain or lodged canopy conditions, delayed or inaccurate control may lead to missed cutting, excessive straw intake, or increased feeding load.

Plant height detection mainly relies on non-contact 3D sensing technologies, including LiDAR, RGB-D systems, structured light cameras, and UAV-based imaging. These devices provide spatial crop information for height estimation and modeling. Common methods include maximum height difference, average height within local windows, and main stem fitting for structured crops such as corn and sorghum[40]. For example, stereo 3D imaging combined with tilt compensation reduced measurement error from 15.6 to 3.9 cm through coordinate transformation, as illustrated in Figure 3[41]. Centimeter-level measurement accuracy is generally sufficient to support header height adjustment and basic operational control during harvesting. However, under complex uneven terrain, its effectiveness in preventing header damage is still influenced by terrain variation and equipment tolerance. Therefore, adaptive control strategies are required to ensure operational safety.

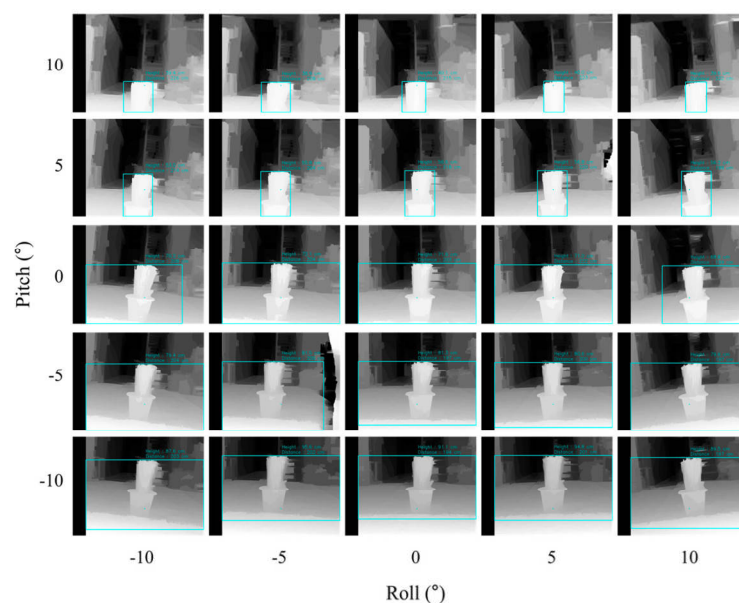


Figure 3. Results of crop height measurement. Reproduced from Kim et al.[41]; licensed under CC BY 4.0.

In field applications, crop height estimation often requires preprocessing to reduce terrain-related noise and improve measurement stability. A common strategy is to derive crop height maps (CHMs) by comparing the crop surface represented by digital surface models (DSMs) with the ground reference described by digital terrain models (DTMs). Point cloud denoising, filtering, and local surface reconstruction can then be used to obtain more reliable height information. In addition, temporal image sequences and semantic crop masks can be integrated to maintain spatial-temporal consistency and to distinguish crop pixels from background regions during height estimation. For example, Li combined UAV-based multisource sensing, 3D reconstruction, and machine learning to estimate winter wheat biomass with high accuracy[42].

In summary, plant height is an important basis for adjusting harvesting machinery parameters, directly affecting path planning and control strategies. Methods based on 3D reconstruction and structural feature extraction have become important perception approaches in intelligent harvesting systems. By providing height information directly related to header height adjustment and feed rate control, these sensing and analysis methods help transform plant height from a crop growth indicator into an operational parameter for harvesting control. However, terrain variation, canopy occlusion, and mechanical vibration still introduce measurement uncertainty and reduce system robustness, while trade-offs among accuracy, cost, and real-time performance across sensing modalities limit stable application.

3.1.2. Plant Density

Plant density describes the population size of crops within a given field area and is generally quantified by the number of plants per square meter. In agricultural standards, this attribute is usually reported using the unit “plants·m⁻²”. It provides a basic indicator for characterizing planting patterns, crop population development, and yield-related field conditions. In practice, it can be measured using fixed quadrat counting or row and plant spacing estimation methods[43]. Accurate knowledge of field crop density distribution is fundamental for operation path planning, feed control, and harvesting strategy formulation in mechanical harvesting. Operation efficiency, machinery load management, and yield estimation are greatly influenced by such information.

Plant density information provides an important basis for regulating harvester operating parameters. When crop stands are dense, the amount of material entering the machine increases, which may cause feeding congestion, insufficient threshing, and greater grain damage. In areas with lower plant density, the forward speed can be increased within an appropriate range to maintain harvesting efficiency and reduce unnecessary fuel consumption. Moreover, density distribution is closely associated with lodging tendency, spike spatial arrangement, and nutrient utilization, and can therefore support pre-harvest field evaluation and the preliminary setting of machinery parameters[44]. During continuous harvesting, density changes along the travel path need to be detected with sufficient real-time performance so that forward speed and feed rate can be adjusted before the crop flow entering the threshing unit fluctuates sharply. Meanwhile, the robustness of density estimation under plant overlap and dusty conditions is also important, because delayed or unreliable speed regulation may increase the risk of local overfeeding, blockage, or unstable threshing load.

Plant density detection mainly relies on image-based counting and point cloud estimation using sensors such as RGB cameras, RGB-D cameras, and LiDAR. Object detection and segmentation models (e.g., YOLO and CenterNet) are effective in low- to medium-density fields, while density regression methods (e.g., CSNet and MCNN) handle dense and occluded conditions by modeling object distributions[45]. For example, a UAV-based

model (DeNet) incorporating global attention and heatmap fusion achieved superior performance to SegNet and U-Net in wheat density estimation, as illustrated in Figure 4[46].

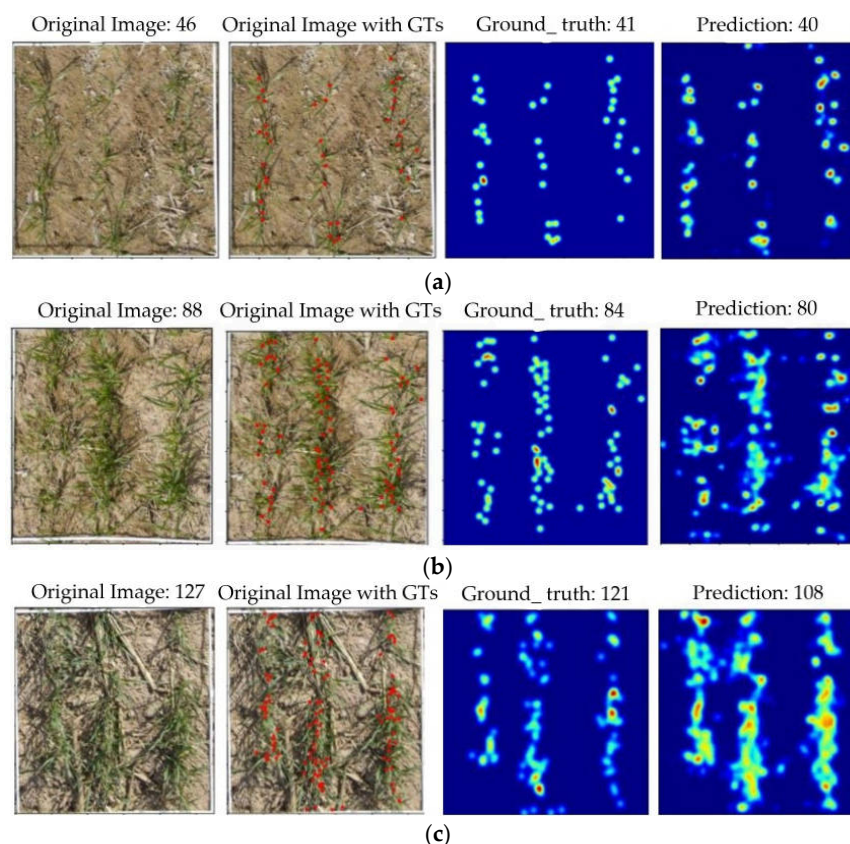


Figure 4. Examples of wheat plant density estimation. (a) low; (b) moderate; (c) high plant density levels. Reproduced from Peng et al.[46]; licensed under CC BY 4.0.

For crops with clear vertical and structural characteristics, such as maize and sorghum, LiDAR-derived point clouds provide an effective data source for density estimation. In these methods, plant population density is usually estimated by extracting spatial clusters from the point cloud, evaluating local point-density patterns, or calculating projected point distributions within selected ground units[47]. The resulting density information can also be visualized as prescription maps or distribution histograms, which helps support harvester scheduling and analysis of within-field variability.

Under field harvesting conditions, plant distribution is rarely uniform, and the apparent density of the crop population may change with growth stage and local growth status. Crop-specific tillering or branching patterns can further obscure individual plants, making density recognition more difficult. To improve estimation stability, recent studies have incorporated temporal observations, semantic-level crop segmentation, and multi-modal sensing information into density estimation frameworks. In densely planted areas, geometry-based separation of plant clusters has also been used to separate neighboring individuals and reduce counting errors[48]. Furthermore, dust generated by combine harvesters reduces image contrast and introduces scattering noise, thereby weakening target boundaries and texture features. For detection-based methods, this may lead to incomplete target features, resulting in missed or false detections. For density map-based methods, dust interference may affect the learning of spatial distribution features, thereby reducing regression accuracy. Therefore, dusty environments pose a significant challenge to the robustness of density estimation methods.

In summary, plant density is one of the core attributes affecting harvesting operation strategies. Detection methods combining image recognition and point cloud density

analysis can efficiently model spatial distribution and support machinery decision-making. The density information obtained can further assist harvesting machinery in adjusting forward speed, feed rate, and regional operation strategies according to crop distribution. However, in practical harvesting environments, occlusion, plant overlap, and dust interference can reduce detection accuracy and system robustness. Meanwhile, image-based methods and point cloud analysis involve trade-offs in computational efficiency, environmental sensitivity, and real-time performance, limiting their stable application in high-speed harvesting scenarios.

3.1.3. Spike Number

Spike number, referring to the number of spikes per unit area or sample region in crops such as wheat and rice, serves as a key indicator for assessing yield potential and harvesting complexity. According to agricultural standards, spike number is obtained by manually counting within a designated sample area (e.g., 0.5 m × 0.5 m), and the result is expressed as “spikes·m⁻²” [38]. Spike number not only determines the upper limit for yield estimation but also provides guidance for controlling the feed rate of combine harvesters, adjusting header load, and configuring the threshing system.

During the harvesting stage, accurately identifying spike number helps intelligent agricultural machinery dynamically adjust travel speed and threshing intensity. Accurate identification of spike number prevents clogging, damage, or grain loss caused by over-feeding spikes. Additionally, spike distribution affects header path optimization and the integrity of harvesting operations. Missing spikes directly result in yield loss, while overlapping areas may lead to crop re-harvesting and increased energy consumption. Therefore, precise perception of spike number is a crucial foundation for ensuring harvesting efficiency and quality control [49]. In practical operations, robust spike counting under occlusion, overlap, and machine-induced image blur helps identify locally sparse or dense grain-bearing regions. This information can support header coverage adjustment and reduce missed harvesting or repeated cutting in areas with uneven spike distribution.

Spike detection mainly uses high-resolution imaging (RGB, UAV, LiDAR-based 3D) combined with deep-learning detectors (e.g., YOLOv5, Faster R-CNN, and SSD) for automatic recognition and counting. These methods can distinguish spikes from leaves, weeds, and other non-target objects in complex backgrounds. An example of wheat spike detection results is shown in Figure 5, demonstrating good generalizability and accuracy. For instance, Zhang proposed an improved YOLOv5s-T method, which efficiently detects wheat spikes based on a multi-growth-stage dataset. The R² values for three periods were 0.97, 0.85, and 0.78, with an overall R² of 0.87. The model size was reduced by 36.8%, and inference efficiency was improved [50].

Due to occlusion, overlap, and pose variation, 2D image-based spike detection may lead to missed or duplicate counts. To address this, methods such as density regression (e.g., CSRNet), multi-scale feature extraction, and instance segmentation (e.g., Mask R-CNN) improve recognition in dense scenes [51]. In addition, combining image data with depth information (e.g., RGB-D or point clouds) enables 3D spike localization and multi-view fusion, enhancing counting accuracy and robustness. Spike counting is typically performed using direct counting or regression-based methods: the former relies on explicit detection or segmentation, while the latter estimates total counts from image features, making it suitable for dense or occluded conditions [52]. Some systems also generate spike distribution heatmaps to support path planning and feed rate control.



Figure 5. Wheat spikes count.

In summary, spike number, as a core indicator of crop yield and harvesting complexity, plays an important role in intelligent agricultural machinery scheduling. The integration of object detection, density estimation, and 3D vision enables efficient spike recognition and counting. The obtained spike distribution and counting results can provide a basis for feed rate regulation, harvesting load assessment, and local operation adjustment. However, in dense and dynamic field environments, severe occlusion, spike overlap, and morphological variation can easily lead to missed detections and counting bias. Detection-based methods rely on clear target boundaries and tend to degrade in dense scenes, whereas regression-based methods are more robust to occlusion but still limited in spatial localization and local accuracy.

3.1.4. Crop Lodging

Crop lodging refers to the phenomenon where, in the later stages of growth, crop stems deviate from the upright growth direction and fall to the ground due to factors such as wind, rain, or unstable physiological structure. According to national standards, the degree of crop lodging is typically characterized by the angle between the stem and the ground[53]. Lodging not only affects the normal maturation of crops and their ventilation and light conditions, but also poses a serious threat to the operational stability and harvesting efficiency of machinery.

Crop lodging can cause spikes to be covered by stems and leaves or distributed close to the ground, leading to missed cutting, excessive cutting height, or spike damage. Lodging typically occurs in blocky or irregular patterns. If the path is not adjusted in time, it can result in repetitive operations, increased energy consumption, and uneven threshing loads[54]. Therefore, early identification of lodging areas and their distribution is a key prerequisite for ensuring stable operation and precise decision-making of intelligent harvesting machinery. In lodged fields, the robustness of lodging-area and lodging-direction detection is particularly important because crop posture can change rapidly with local terrain, stem orientation, and canopy overlap. Reliable lodging information can support timely path correction and header attitude adjustment, enabling the cutter to approach low-lying spikes more effectively and reducing missed cutting or stem accumulation at the header.

Lodging detection primarily relies on visual and 3D perception technologies. Commonly used sensors include RGB cameras, LiDAR, structured light depth cameras, and UAV-based multi-angle imaging systems. RGB images combined with deep segmentation models (such as U-Net and SegFormer) can achieve pixel-level localization of lodging

areas[55]. The detection results of the rice lodging areas are shown in Figure 6. LiDAR generates canopy point clouds, and ground-level lodging can be identified by combining height models, ground fitting, and anomaly analysis. Furthermore, methods such as angle calculation, directional statistics, and spatial clustering can be used to quantitatively analyze lodging angles and directions[56].

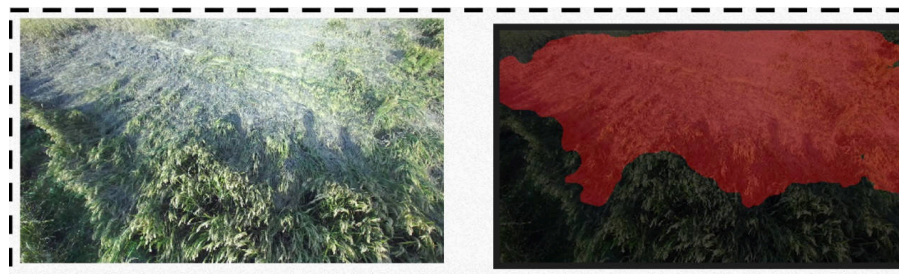


Figure 6. Detection results of rice lodging.

Point clouds provide high-precision estimation of lodging angles by fitting stem vectors to ground plane normals. In addition, texture-based methods (e.g., Fourier and Gabor filters) analyze RGB images to determine lodging direction. Combining sliding window and projection techniques enables the generation of lodging distribution maps for path planning and header adjustment. For example, a depth camera-based method extracted lodging location, severity, and direction with errors ≤ 9.0 cm in severity, $\leq 9.0^\circ$ in direction, and ≤ 5.0 cm in harvesting height adjustment[57].

In summary, crop lodging directly affects harvesting success rate and operational efficiency, making it a key attribute in intelligent harvesting control. The integration of point cloud modeling, image segmentation, and posture analysis improves lodging detection accuracy and spatial representation. Information such as lodging area, direction, and severity can further support header height adjustment, travel path optimization, and forward speed control in lodged fields. However, lodging patterns are irregular and spatially heterogeneous, and are easily affected by occlusion, ground interference, and posture variation, leading to limited detection stability. Image-based methods are sensitive to illumination and background complexity, while point cloud methods offer structural representation advantages but depend on data quality and involve high computational cost; posture estimation may also produce errors under severe lodging or mixed conditions.

3.1.5. Canopy Structure

Canopy structure describes the vertical and horizontal spatial distribution of crop populations, reflecting characteristics such as density, layering, leaf area, and shading, and is typically quantified by parameters including height, leaf area index, coverage, and porosity[58]. It influences photosynthesis, ventilation, and lodging resistance, and also affects harvesting operations such as cutting resistance, path planning, and crop flow. In complex or dense canopies, interlaced layers and occlusion can increase cutting load and lead to missed harvesting, reducing image recognition accuracy and feeding efficiency[59]. Accurate canopy structure characterization is therefore essential for optimizing path planning, adaptive control, and target detection.

Canopy structure, as an important attribute reflecting the spatial distribution characteristics of crops, exhibits significant differences across crop types. For grain crops, canopy structure is generally more uniform, and related studies mainly focus on population height distribution and coverage. In contrast, for fruit and vegetable crops, canopy structure is more complex and is often influenced by factors such as branch and leaf distribution, fruit position, and growth stage. Therefore, in canopy structure detection, different

crop types often require differentiated modeling and feature extraction methods, which increases the complexity and diversity of related research. For harvesting machinery, reliable canopy-structure information can help estimate cutting resistance, crop-flow continuity, and potential occlusion before the machine reaches dense or layered crop regions. However, because detailed canopy reconstruction usually requires depth sensing or multi-view data, its practical application still needs to balance detection robustness, computational load, and sensor deployment cost.

Canopy structure perception typically relies on LiDAR, RGB-D cameras, UAV multi-view imaging systems, and near-ground remote sensing platforms. Current mainstream methods include point cloud-based 3D modeling and analysis, image-based texture and cover estimation, and multispectral imaging for extracting vegetation indices[60]. For example, Xie proposed a method for extracting potato canopy parameters based on UAV RGB images, with the extracted canopy structure point cloud shown in Figure 7. Using vegetation indices such as EXG to estimate canopy coverage, the EXG index showed the best performance with an R^2 of 0.94. Combining coverage with pesticide application models, accurate variable-rate spraying in the field was achieved[61].

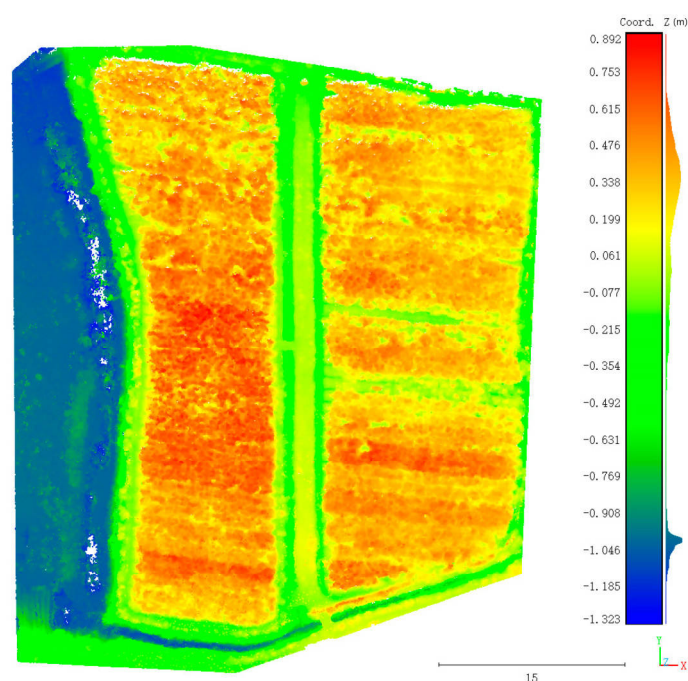


Figure 7. Point cloud of canopy height. Reproduced from Xie et al. [61]; licensed under CC BY 4.0.

In point cloud analysis, by dividing the 3D point cloud obtained from LiDAR scanning into grids and performing elevation fitting, the upper and lower boundaries of the canopy and its thickness information can be extracted. The coefficient of variation (CV) of canopy height is then calculated to reflect its spatial uniformity. Furthermore, by calculating the point density distribution per unit area and the point cloud intensity gradient in the vertical direction, the canopy layer structure and porosity can be estimated. Assessment of light penetration conditions and mechanical cutting resistance can be achieved based on the obtained information.

Image-based methods typically rely on low-altitude UAV RGB images, combined with convolutional neural networks (such as SegNet, DeepLabv3+) for canopy segmentation and canopy cover estimation. Canopy distribution analysis over large areas can be effectively performed in this way, enabling operational strategy adjustment at the regional scale[62].

In summary, canopy structure is an important spatial feature reflecting crop population status and significantly affects path planning, header control, and crop flow. The integration of multi-source sensing information with point cloud modeling and image segmentation enables canopy structure reconstruction and analysis. The reconstructed canopy structure can provide information for crop flow estimation, header operation adjustment, and harvesting path optimization under dense or uneven crop growth conditions. However, layered occlusion, uneven density, and complex vertical structures can lead to discontinuous segmentation and errors in structural parameter estimation. Image-based methods are sensitive to illumination and background variation, while point cloud methods offer advantages in 3D representation but depend on data completeness and involve high computational cost; registration errors in multi-sensor fusion may also affect canopy representation accuracy.

3.1.6. Crop Position

Crop position refers to the spatial coordinates of individual crops or crop populations, typically including two-dimensional (X, Y) and, in some cases, vertical (Z) information. It is a fundamental attribute for precision agricultural operations, particularly in combine harvesting, autonomous navigation, and zoned management[63]. Crop position detection supports spatial mapping as well as tasks such as path planning, header control, row alignment, and localized operation adjustment.

Crop position detection is implemented on two levels in practical applications: At the macro scale, “distribution map” modeling analyzes the spatial distribution of crops in the field to generate density heatmaps, identify vacant areas, and perform regional variability analysis. At the micro scale, “row detection” identifies the centerline and boundaries of planted rows in crops such as corn, wheat, and rice, thereby supporting autonomous navigation, header alignment, and inter-plant positioning. For harvesting machinery, position information needs to be updated with sufficient real-time performance to correct navigation deviations during machine movement and maintain header alignment with crop rows. Meanwhile, the robustness of crop-row or crop-position detection under missing plants, irregular planting, temporary GNSS instability, or visual occlusion is also important, as it determines whether perception results can be effectively integrated with steering control and local operation planning.

Distribution maps are generated by fusing GNSS positioning with multi-source remote sensing data. Crop images or point clouds are aligned with geographic coordinates to produce heatmaps or vegetation index maps representing crop density, coverage, and health. These methods can identify empty sowing, missing seedlings, or re-sowing areas, supporting adaptive feed rate control and zonal harvesting strategies[64].

Row detection is used in path control for combine harvesters and autonomous platforms, with methods based on images or LiDAR point clouds. Image-based approaches use edge detection (e.g., Canny and Hough) to extract row gaps and centerlines, while LiDAR methods model crop rows using point cloud density and spatial features, adapting to complex lighting and occlusion. For example, a LiDAR-based row centerline detection method combining improved K-means and least squares fitting achieved 95.1% accuracy, with the detection results shown in Figure 8[65].

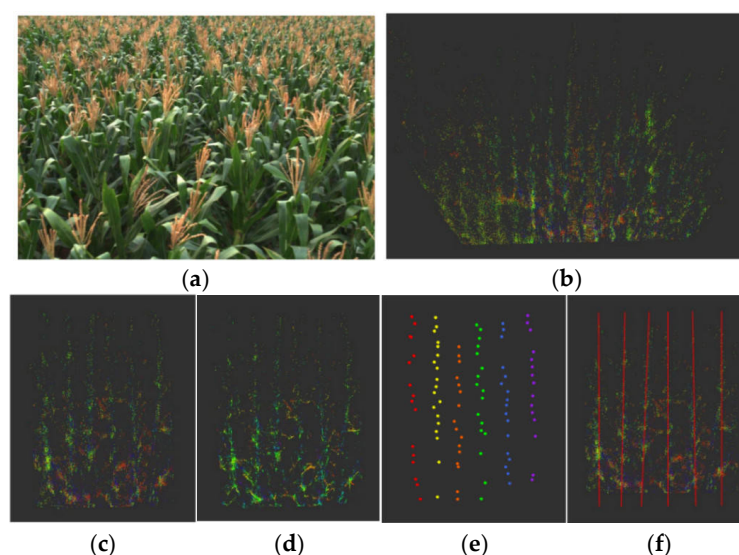


Figure 8. Crop positions and row centerline. (a) the environment of the late period maize field; (b) the point clouds of late period maize; (c) enlarged image of detection area; (d) denoising using Gaussian filtering; (e) feature points of late period maize; (f) the centerlines of late maize rows. Reproduced from Zhang et al.[65]; licensed under CC BY 4.0.

In summary, crop position detection is fundamental to spatial digitization of agricultural operations and precise machinery control. By integrating multi-source spatial information with image processing, point cloud analysis, and coordinate calibration, crop distribution maps can be constructed and row structures identified, supporting efficient and low-loss harvesting. Such spatial information can further guide row alignment, navigation correction, header positioning, and local operation adjustment during harvesting. However, occlusion, irregular planting, and GNSS signal instability can affect positioning accuracy and introduce cumulative errors during mapping and alignment. Image-based row detection is susceptible to background interference, point cloud methods require high data quality and computational resources, and registration errors in coordinate transformation and multi-sensor fusion may also affect spatial consistency.

3.1.7. Other Attributes

In cereal crop harvesting, conventional attributes include plant height, density, and spike number. Some finer, less-studied attributes also significantly influence system performance and operational precision under specific harvesting conditions. These attributes mainly include spike layer height, stem diameter, spike pedicel length, stem stiffness, and lateral branch distribution characteristics. Such information supports cutter position control, machine path adjustment, threshing strategy formulation, and loss reduction during harvesting.

Spike layer height refers to the relative height distribution of the crop spikes within the population, typically characterized by measuring the distance between the top of the spike and the ground. Accurate knowledge of spike layer height is crucial for guiding cutter height settings, especially in combine harvesters with a dual-cutter structure. The mean and variance of the spike layer can be used to dynamically adjust the position of the upper cutter, enabling efficient, low-loss harvesting of spikes. The approach is particularly suitable for direct grain harvesting and high-density planting models[66].

Stem diameter is an important parameter affecting threshing efficiency and machine energy consumption. Thicker stems often lead to increased feeding resistance, higher threshing load, and greater machine wear and fuel consumption. Recent studies have explored indirect estimation of stem size and stiffness using LiDAR cross-sectional analysis,

RGB-D depth map fitting, or flexible tactile sensing, enabling pre-setting and adaptive adjustment of threshing drum parameters. For example, Zhou proposed a corn stem monitoring technology based on the internal gradient algorithm, integrating HSV analysis and morphological operations to obtain stem diameter. The detection result, shown in Figure 9, has a measurement error of 1.92 mm and performs stably in field conditions[67].

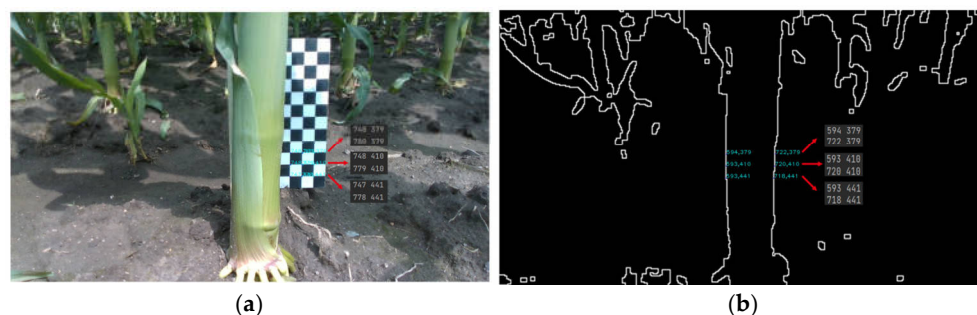


Figure 9. Corn stem diameter. (a) coordinate extraction of the checkerboard; (b) coordinate extraction of the maize stem contour. Reproduced from Zhou et al.[67]; licensed under CC BY 4.0.

Spike pedicel length refers to the connecting distance between the spike and the main stem in crops such as rice and wheat. The drooping angle and visibility of the spike are affected by pedicel length, which in turn influences target occlusion and detection accuracy in image recognition. Accurate estimation of spike pedicel length helps assess the ease of spike harvesting and supports the optimization of target segmentation algorithms to improve spike detection accuracy[68].

In addition, the distribution of lateral branches and the number of tillers hold certain value for planning the harvesting path of combine harvesters and evaluating canopy structure. A high number and dense distribution of lateral branches can improve the harvestable rate of crops, but also increase leaf occlusion and impurity content. Therefore, in fields with complex canopy structures, sensing the morphology of lateral branches and tillers can help optimize harvesting paths, header speed, and wind separation intensity. For example, Wu proposed a wheat tiller number prediction method that integrates gradient features and vegetation indices, achieving an R^2 of 0.7044–0.7357 and improving the prediction accuracy of wheat tiller number by 32.6–41.8%[69].

Compared with mainstream attributes such as plant height, plant density, and lodging, which directly affect harvesting efficiency, operational parameter adjustment, and loss control, attributes such as spike layer height, stem diameter, and spike pedicel length have a relatively limited influence on the basic setting of harvesting operations and intelligent operation control[40]. Nevertheless, these attributes can still provide auxiliary supporting information for operational decision-making and control. Current research on these attributes usually relies on high-resolution imaging, detailed 3D point cloud modeling, and local structural feature extraction, which impose high requirements on calibration accuracy and real-time computation. However, due to limited research attention, annotated datasets for these attributes are generally scarce and lack unified standards, restricting model training and generalization. Meanwhile, fine-grained feature extraction also increases computational complexity, posing challenges for real-time application on embedded harvesting platforms.

3.2. Attributes of Fruit and Vegetable Crops

Fruit and vegetable harvesting differs from grain crop harvesting, as the perception targets are usually individual objects with complex spatial distributions, diverse morphologies, and asynchronous maturity. Therefore, the perception system needs to identify key

attributes such as maturity, three-dimensional position, integrity, and health status to support harvest timing judgment, target localization, path planning, and operation decision-making. This section focuses on maturity, position, quality, and additional features such as geometric anomalies and connection structures.

3.2.1. Maturity

For fruit and vegetable crops, maturity is closely associated with the appropriate harvest window, postharvest storability, transport tolerance, and commercial grading. In agricultural standards, maturity is generally described from several perspectives, including external appearance, physiological development, and harvest suitability. Its evaluation usually involves both external and internal indicators, such as peel color, surface texture, water status, soluble solids, starch accumulation, protein content, and acidity[70]. Because the ripening process differs among crop species, the criteria used for maturity assessment are also crop-specific. For example, citrus maturity is commonly judged by color development and the sugar–acid balance, whereas banana maturity is more closely related to the conversion of starch into sugars.

In automated harvesting, maturity information directly affects harvest timing, target selection, and batch operation scheduling. Fruits or vegetables that have not reached the required maturity may have low commercial value or poor postharvest quality, whereas overmature targets are more susceptible to mechanical damage during picking and transportation. Therefore, maturity detection can provide decision support for selective harvesting, picking sequence planning, and pre- or post-harvest grading. In practical deployment, maturity recognition should meet the real-time requirements of online target selection, so that the picking system can determine whether a target should be harvested before the end-effector approaches it. Meanwhile, the robustness of maturity assessment under illumination variation, fruit occlusion, and cultivar differences is also important, because unstable classification results may lead to missed harvesting, incorrect picking decisions, or inconsistent grading. In addition, maturity detection results need to be integrated with picking-path planning, end-effector control, and grading systems to support continuous and selective harvesting operations.

Maturity assessment of fruits and vegetables usually requires the combined use of multiple sensing modalities to characterize both surface appearance and internal quality changes. Among external traits, peel color provides a direct and easily observable cue for ripeness evaluation. During ripening, changes in pigment composition alter the visual appearance and spectral reflectance of the fruit surface. These variations can be described using different color representations, including RGB, HSV, and Lab spaces. For instance, banana ripening can be reflected by changes in the Lab *b* channel during the green-to-yellow transition, whereas tomato maturity is often evaluated using redness-related indices. To obtain more detailed spectral information, multispectral and hyperspectral imaging systems are used to record reflectance responses over a broad wavelength range from the near-ultraviolet to the near-infrared region. Based on these data, indices such as NDVI, GCI, and RVI can be calculated to improve the description of ripening-related physiological changes[71].

During ripening, fruit and vegetable crops undergo changes in several internal constituents, including water, soluble sugars, starch, organic acids, proteins, and volatile compounds. Spectral sensing techniques, especially near-infrared spectroscopy (NIR) and hyperspectral imaging (HSI), are frequently used for non-destructive assessment of these changes. These methods infer maturity status by capturing wavelength-dependent responses associated with water, sugars, lipids, and other biochemical components. For example, Zhao applied near-infrared hyperspectral imaging together with deep learning to estimate tomato quality traits and identify ripeness categories, as illustrated in Figure 10.

Among the tested models, the RNN-based model showed the best classification performance, and the R^2 values for the predicted quality parameters were all higher than 0.87, showing better performance than conventional methods[72].

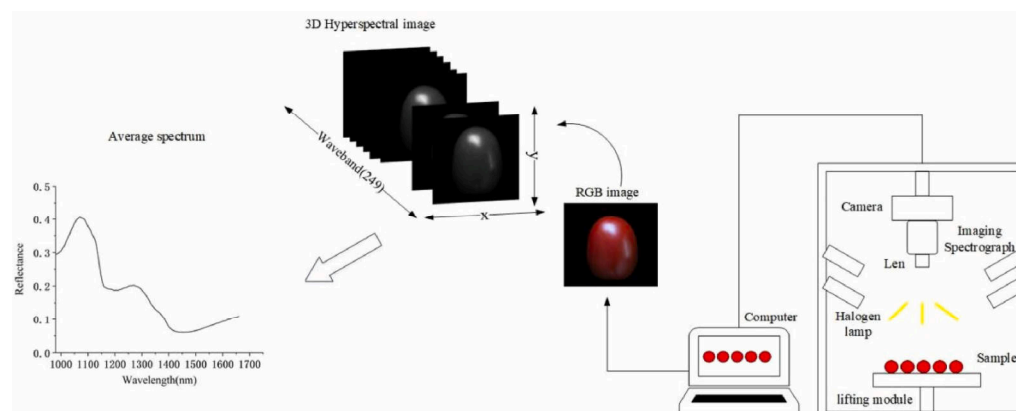


Figure 10. Hyperspectral imaging workflow for tomato maturity detection. Reproduced from Zhao et al.[72]; licensed under CC BY 4.0.

For more detailed analysis of biochemical changes during ripening, attention can also be given to processes such as starch degradation, reducing sugar formation, and volatile compound release. These changes can be investigated using Raman spectroscopy, Fourier-transform infrared spectroscopy (FTIR), and dielectric-based measurements. Raman spectroscopy is suitable for detecting molecular vibration signals associated with organic constituents, including starch and pectin, and is therefore useful for tracking sugar-related transformations in fruits such as kiwifruit and apples. In comparison, FTIR can be used to characterize absorption features related to moisture and proteins, with representative bands commonly appearing near 1640 cm^{-1} and 1530 cm^{-1} , respectively[73].

In summary, accurate detection of fruit and vegetable maturity is important for determining optimal harvest timing, improving operational efficiency, and enabling reliable quality grading. By integrating multi-source information such as color, chemical composition, and physiological status, current methods can comprehensively characterize maturity. The maturity information obtained can support selective picking, batch operation scheduling, and grading decisions in automated fruit and vegetable harvesting systems. However, differences in crop varieties and environmental conditions may introduce detection uncertainty and limit cross-scenario generalization. Meanwhile, the trade-off between detection accuracy and real-time performance remains a key constraint on its application in harvesting systems.

3.2.2. Position

For fruit and vegetable harvesting, spatial perception provides essential information for determining picking routes, controlling end-effectors, and selecting harvestable targets. This information involves two spatial scales. At the field or plant level, it describes crop arrangement patterns, including row spacing, plant spacing, and planting density. At the fruit level, it specifies the three-dimensional location and pose-related features of individual fruits, such as height, lateral displacement, and the relative angle between the fruit and the main stem[74]. In automated picking systems, reliable spatial perception enables the machine to avoid obstacles, approach targets accurately, and complete picking operations with reduced mechanical damage.

In plant position perception, the goal is to obtain the row structure and field distribution of crops, supporting path planning and operational optimization. For example, in orchards, identifying row spacing, plant spacing, and tree arrangement helps achieve

high-precision row navigation, reducing deviations and collisions. In greenhouse and similar environments, plant distribution information aids in block division and path optimization. Commonly used sensors include RGB-D cameras, LiDAR, and multi-view imaging platforms. RGB-D devices, due to their ability to simultaneously capture color and depth information, are widely used for point cloud modeling and spatial heat map generation. Multi-source perception methods integrating GNSS, IMU, and semantic segmentation have achieved sub-centimeter plant positioning accuracy[75]. In practical harvesting systems, plant-position information needs to be updated with sufficient real-time performance so that the mobile platform can adjust its navigation path before entering narrow rows, turning areas, or densely planted blocks. Meanwhile, the robustness of position perception under branch and leaf occlusion, irregular plant spacing, and accumulated positioning errors is also important, as it determines whether perception results can be effectively integrated with platform navigation, block-level path planning, and subsequent robotic arm operations.

The calculation of fruit position typically involves the following steps: firstly, object detection or instance segmentation is used to obtain the two-dimensional position of the fruit in the image. Then, by combining depth maps or point cloud data, its three-dimensional coordinates are calculated to determine its exact location, as shown in Figure 11. Common methods for depth acquisition include disparity calculation, laser ranging, and structured light[76]. To achieve fruit localization in real space, camera calibration and coordinate transformation are required to convert image coordinates to the robot world coordinate system. To enhance the robustness of positioning in complex environments, research often integrates multi-view fusion, SLAM, or visual-inertial odometry (VIO) techniques for spatial reconstruction[77]. For occluded or overlapping fruits, three-dimensional pose estimation and keypoint regression can also be used to assist in determining their spatial orientation and harvestability.

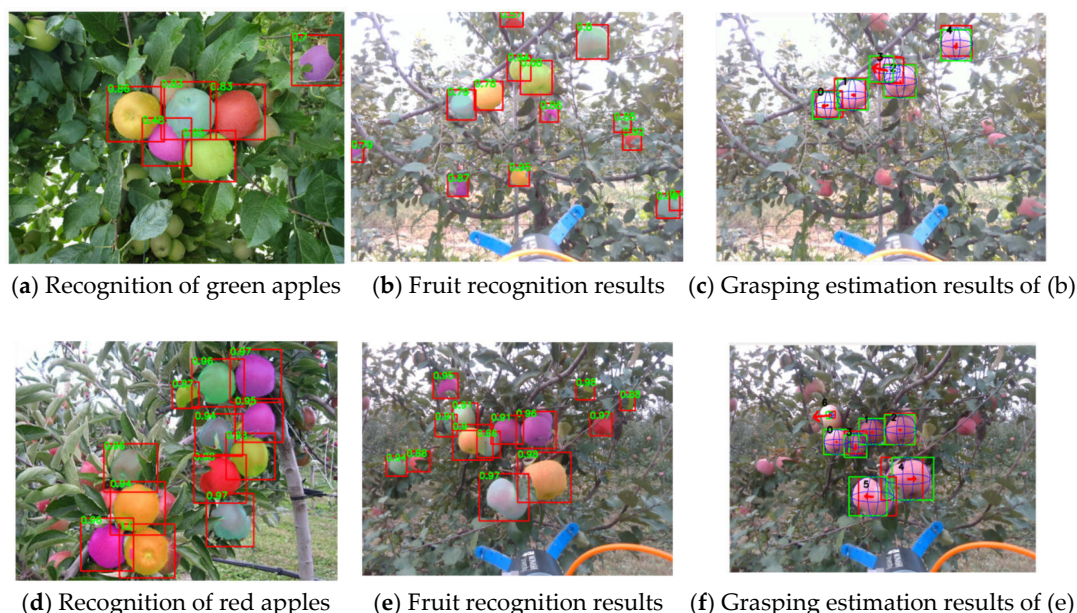


Figure 11. Fruit locations. Reproduced from Kang et al.[76]; licensed under CC BY 4.0.

In summary, position attribute detection of fruit and vegetable crops should be based on a two-layer “plant–fruit” structure for hierarchical modeling and collaborative perception. This approach ensures the spatial rationality of platform-level paths and operation strategies while supporting precise localization and motion planning of end-effectors. By combining plant-level spatial distribution information with fruit-level 3D localization

results, position detection can provide key information for navigation, target selection, picking path planning, and end-effector control. However, in practical environments, occlusion between fruits and foliage, plant structural differences, and viewpoint limitations may affect positioning accuracy and continuity. Meanwhile, multi-source data fusion and hierarchical perception still face challenges in data alignment and computational efficiency, constraining their stable application in real-time systems.

3.2.3. Quality

Crop quality is one of the key attributes that need to be detected during fruit and vegetable harvesting, directly determining the market value and storage performance of agricultural products. In fruit and vegetable crops, quality mainly refers to whether the fruit is healthy and intact, and whether there is any infestation by pests and diseases, mechanical damage, or softening and rotting. Quality issues typically exhibit diversity, locality, and irregularity, with uneven distribution and visual signs such as subtle color spots, deformation, or texture changes. Therefore, the precision and robustness of the perception system in automated harvesting systems are subject to higher demands.

Crop quality is crucial because damaged or diseased fruits reduce market value and can promote rot during transportation and storage, causing greater losses. In automated harvesting, if these issues are not identified in time, it will increase sorting pressure and interfere with harvesting strategies (such as misjudging the picking points). Furthermore, pest and disease distribution often exhibits certain spatiotemporal correlations, and timely monitoring can provide feedback for orchard pest and disease control strategies[78]. For fruit and vegetable crops, quality attributes usually involve multiple aspects, including sugar content, firmness, moisture content, and internal tissue structure, which are often indirectly characterized using spectral or near-infrared technologies[79,80]. Significant differences exist among crops in terms of quality evaluation criteria and detection methods. For example, fruit quality detection typically focuses on internal maturity and soluble solids content, whereas vegetable quality assessment emphasizes external appearance and freshness[81–83]. Therefore, in practical applications, quality detection methods need to be specifically designed according to the crop type, which further increases the complexity of related research. For online harvesting decisions, quality assessment needs to provide sufficiently timely results so that defective, diseased, or unsuitable targets can be avoided or separately handled before picking. Meanwhile, the robustness of quality detection under uneven illumination, partial occlusion, and subtle early-stage defects is also important, as it determines whether perception results can be effectively integrated with selective picking, quality grading, and post-harvest sorting systems.

Quality perception typically relies on multimodal visual features, with color changes, such as dark or discolored areas, and rough edges serving as direct indicators of lesions or damage, making RGB images commonly used. However, visible light information can sometimes fail to detect early-stage diseases, so Near Infrared (NIR), thermal imaging, and hyperspectral imaging (HSI) sensors are gradually being introduced. NIR analyzes sugar and moisture changes, thermal imaging detects temperature anomalies, and hyperspectral imaging captures lesion reflectance across multiple bands, enabling early and comprehensive quality assessment[84].

Traditional quality analysis relies on statistical features such as GLCM and LBP combined with classifiers like SVM and RF. In recent years, deep learning has become the mainstream approach, with CNN-based models such as ResNet, EfficientNet, and U-Net achieving high-precision detection of surface defects, including lesions, rot, and damage. Instance segmentation further enables precise annotation of defect regions, supporting harvestability assessment and multi-level sorting. For example, a YOLOv4-based guava quality detection system achieved 88.15% accuracy, with misdetection and missed

detection rates below 9% and real-time performance at 12 FPS. The detection results are shown in Figure 12[85].

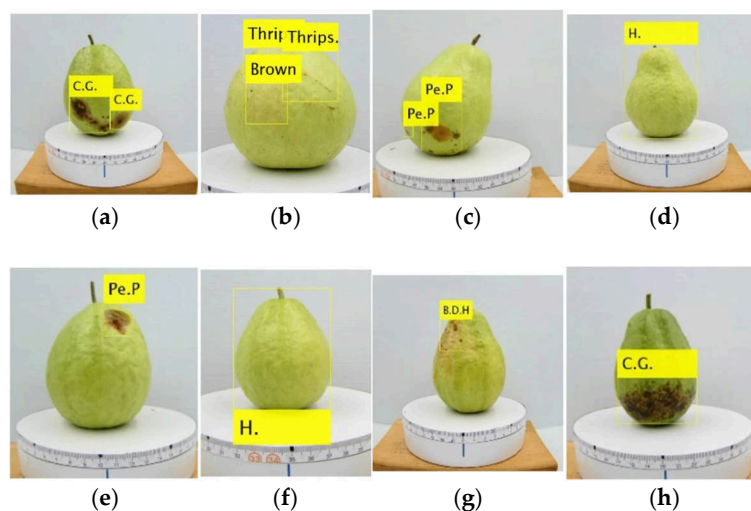


Figure 12. Guava quality. (a) training image 1; (b) training image 2; (c) training image 3; (d) training image 4; (e) test image 1; (f) test image 2; (g) test image 3; (h) test image 4. Reproduced from Chiou et al.[85]; licensed under CC BY 4.0.

In summary, fruit and vegetable quality detection plays an important role in assessing crop health and market value within automated harvesting systems. The integration of RGB, near-infrared, and hyperspectral information enables joint characterization of external defects and internal quality attributes. The obtained quality information can further support harvestability assessment, selective picking, and post-harvest grading or sorting decisions. However, quality defects are often subtle and localized, and can be confused with normal color and texture variations under changing illumination. Meanwhile, spectral methods rely on indirect representations, which may introduce uncertainty across varieties and maturity stages; high-accuracy models also increase computational burden and limit real-time application.

3.2.4. Other Attributes

In automated fruit and vegetable harvesting, besides core attributes such as maturity, location, and quality, some less common but important traits exist for specific crops or special scenarios. Such attributes, while not universally applicable, provide auxiliary value for harvest strategy formulation, support mechanical structure optimization, and assist path planning. A valuable complement to the fruit and vegetable harvesting perception system is thus provided by including these attributes.

The atypicality of fruit geometric features is particularly evident in certain special fruit and vegetable varieties, such as misshapen fruits, twin fruits, or non-spherical fruits. Due to the significant differences in morphological structure compared to normal fruits, these fruits often affect gripping stability and transport compatibility. If not detected and identified in advance, mechanical damage or sorting failure may occur. The detection of geometric anomalies mostly relies on 3D vision systems or multi-view image reconstruction technologies, combined with shape descriptors (such as aspect ratio, convex hull area ratio, and curvature distribution) to achieve precise modeling and classification of fruit shapes[86].

The connection structure features between fruits and their stems or between the fruit and the flower receptacle (such as stem-connected, top-hanging, or side-hanging) also

affect harvesting path planning and gripping angle selection. Connection structure features play a particularly important role in multi-fruit vegetables such as tomatoes and eggplants. Different fruit attachment methods often determine the center of gravity and rotational radius of the fruit, imposing varied design and control requirements on the robotic gripper. In terms of detection methods, high-resolution images combined with instance segmentation algorithms and structural boundary modeling are typically used to classify the connection modes. For example, Chu proposed the ALACS 3D positioning system for apples based on laser triangulation, which assists in harvesting, achieving positioning accuracy between 6.9–11.2 mm and a harvesting success rate of 95%[87].

In addition, certain dynamic attributes related to the harvesting rhythm exist in specific crops, such as the natural fruit drop tendency and vibration response characteristics. For example, in vibration-based mechanical harvesting of apple trees, detecting the fruit-branch coupling strength and vibration response frequency helps assess the suitability of non-contact vibration-induced harvesting. Such detection typically relies on accelerometers, miniaturized radar, or high-frame-rate cameras, combined with temporal feature analysis or frequency-domain analysis methods for modeling[88].

In summary, special attributes of fruits and vegetables are often overlooked in traditional detection systems. Although they are not required for all crops and scenarios, they play an important role in specific harvesting tasks. These attributes usually involve irregular morphology, connection structures, or dynamic characteristics, which increase perception complexity and place higher demands on detection accuracy and structural understanding. Meanwhile, limited data support and large crop differences still challenge the generalization and stable extraction of existing methods. Incorporating these fine-grained attributes into system perception can help improve the adaptability and operational precision of harvesting systems.

4. Key Technologies for Crop Attribute Detection

The key technologies for crop attribute detection mainly include sensing technologies for crop data acquisition and data analysis methods for attribute extraction. These technologies determine not only what types of crop information can be obtained, but also whether such information can be accurately and efficiently transformed into attribute results usable for harvesting decisions. Sensors determine the data sources, information dimensions, and perception accuracy, while detection methods extract key attributes from raw images, spectra, or point clouds. Therefore, the selection of sensing technologies and algorithmic models should match specific crop attributes and application tasks. However, for harvesting machinery, technical applicability should not be evaluated only by detection accuracy. It is also necessary to consider whether the system can provide timely outputs during continuous machine movement, maintain stable performance under illumination variation, occlusion, dust, vibration, and uneven terrain, and remain acceptable in terms of sensor cost, computational load, power consumption, and maintenance difficulty. In addition, detection results need to be integrated with onboard control units and mechanical actuators, so that crop attribute information can be converted into practical control actions, such as header height adjustment, forward speed regulation, feed rate control, navigation correction, target localization, and end-effector control. Therefore, the key technologies for crop attribute detection should not only support reliable attribute extraction, but also meet the requirements of real-time performance, robustness, cost, and system integration for practical deployment in harvesting machinery.

4.1. Approaches for Acquiring Crop Data

In agricultural crop harvesting detection, sensors play a key role in obtaining environmental and crop attribute information, determining the dimensionality, accuracy, and stability of data perception. By collecting information such as color, morphology, texture, and spectrum[89], sensors can identify crop growth status, maturity, and quality grades, thereby supporting precise operations and efficient harvesting. Sensor technologies cover two-dimensional visible-light imaging, multi-band fusion, three-dimensional modeling, and internal physiological parameter detection, forming a rich and layered technological system, as shown in Figure 13. This section systematically reviews the main sensing hardware used in agricultural harvesting detection and further analyzes their suitability and application characteristics across different perception tasks, with emphasis on summarizing the functional positioning of different sensing hardware from the perspectives of data acquisition capability and information representation dimensions.

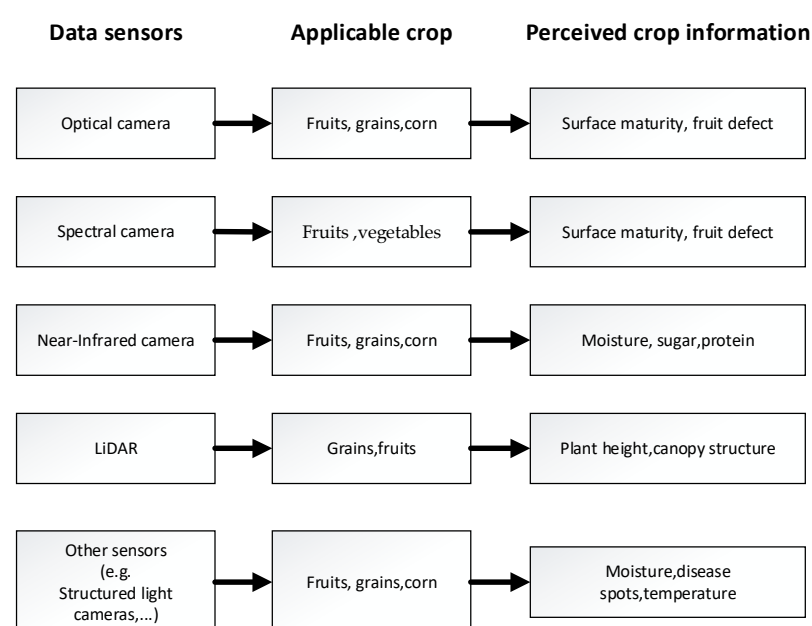


Figure 13. Classification of data sensors.

4.1.1. Optical Camera

The optical camera is one of the earliest and most widely used sensor types in agricultural crop detection. Reflection signals in the red, green, and blue bands are collected and analyzed to generate high-resolution color images. These images characterize basic crop attributes such as appearance, color variation, and morphological structure. As one of the core components of intelligent agricultural vision perception systems, optical cameras serve a wide range of applications, including crop identification, maturity judgment, pest and disease screening, and damage detection. Data acquired from optical cameras provides an important basis for harvesting decisions and operational control[90].

The most common visible light imaging system is the monocular RGB camera, which captures images with a single lens. The simple structure and low cost make the device suitable for most image analysis tasks under standard conditions. Image feature extraction algorithms such as color histograms, texture directions, and edge shapes, combined with classification methods including SVM, K-means, and shallow neural networks, are commonly applied in crop detection. These methods enable preliminary identification of crop types, maturity stages, and surface lesions. In fruit and vegetable sorting lines, a monocular RGB camera can measure fruit size and color for screening, helping the grading system eliminate fruits with skin damage or rot[91]. However, this system has limitations

when facing complex tasks like detecting three-dimensional structures and handling occlusions, as it lacks depth information and sufficient perception dimensions.

To enhance spatial perception capabilities, binocular cameras have been applied in agriculture. Three-dimensional structural information of crops is reconstructed through disparity calculation[92]. Compared to LiDAR, binocular cameras have advantages such as smaller size, lower power consumption, and lower cost, making them suitable for crop height estimation, fruit localization, and path guidance. Studies have shown that combining algorithms like SGBM (Semi-Global Matching) and ELAS (Efficient Large-scale Stereo Matching) can achieve centimeter-level depth estimation, effectively reconstructing the canopy height and spike distribution of crops like corn and wheat.

In recent years, stereo vision cameras have been gradually integrated with traditional monocular systems in agricultural machinery applications. The stereo vision system combines image texture with disparity estimation to generate three-dimensional point cloud data with higher resolution, making the system suitable for fruit recognition and target grasping tasks. For example, in tomato-picking robots, a stereo camera on the end effector enables real-time identification of fruit position, orientation, and maturity, while supporting path planning to improve picking efficiency[93]. In such applications, image acquisition, disparity calculation, and target localization need to be completed before the end-effector reaches the fruit; otherwise, delayed visual feedback may lead to inaccurate grasping points or unnecessary robotic arm adjustments. Therefore, when optical vision systems are used for online harvesting operations, they need to be closely synchronized with robotic motion control.

In summary, optical cameras, as fundamental sensors for crop appearance perception, have evolved from early monocular static imaging to multi-camera, stereo, and dynamic perception systems[94,95]. Their integration with image recognition algorithms and 3D modeling technologies improves the accuracy, adaptability, and real-time performance of agricultural harvesting detection[96]. From an engineering perspective, visible-light vision systems have low cost and convenient integration, making them suitable for large-scale deployment[97,98]. However, they are sensitive to illumination changes and occlusion, and their stability in complex environments still requires further improvement.

4.1.2. Spectral Camera

As the demand for detecting internal physiological states and detailed attributes of crops increases in agricultural harvesting operations, single RGB cameras are no longer sufficient. Multispectral and hyperspectral sensors, which can provide more abundant spectral information, have demonstrated advantages in tasks such as crop health monitoring, maturity assessment, and quality identification. The difference between visible-light cameras and multispectral cameras is illustrated in Figure 14[99].

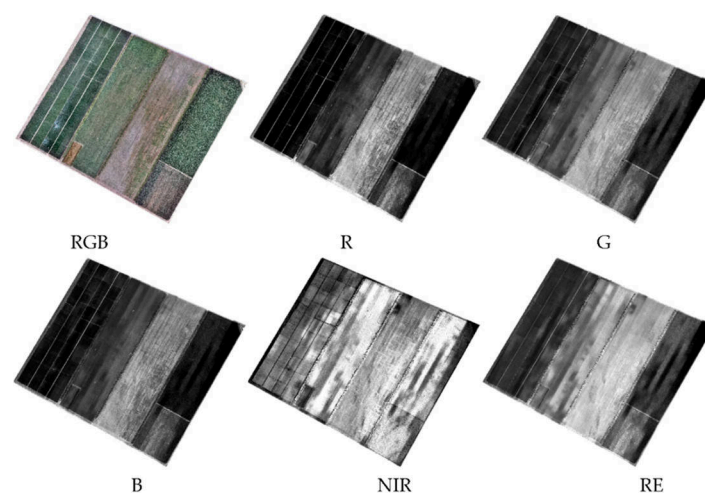


Figure 14. Orthophotos of visible-light and multispectral images. Reproduced from Zheng et al.[99]; licensed under CC BY 4.0.

Multispectral sensors capture crop reflectance changes at different wavelengths by setting several discrete channels (usually 4–12) in the visible and near-infrared bands. Physiological indices such as vegetation index and water index (WI) are extracted from the acquired spectral information. Compared with RGB images, multispectral data offer more precise information on crop chlorophyll, water, and photosynthesis, supporting farmland monitoring, fertilization, and early disease detection[100].

However, multispectral sensors are limited by the number of bands and lack spectral continuity, making it difficult to conduct in-depth detection of internal crop components such as sugars, proteins, and starches. For this purpose, hyperspectral sensors have been introduced into the agricultural detection field. Hyperspectral sensors capture continuous data across tens to hundreds of narrow bands (1–10 nm), allowing pixel-level recording of crop spectral responses for micro-level disease detection, variety identification, and precise maturity assessment[101].

To mitigate the issues of high cost, large data processing volumes, and slow imaging speed associated with hyperspectral devices, recent research has focused on feature band selection and sensor integration optimization. For example, Principal Component Analysis (PCA) and Successive Projection Algorithm (SPA) are applied to extract key bands and construct low-dimensional spectral features, reducing computational complexity and latency while maintaining detection accuracy[102]. In practical harvesting operations, such band reduction is important because spectral decisions related to maturity or disease need to be completed within the operation cycle of fruit selection, grading, or machinery parameter adjustment.

In summary, multispectral and hyperspectral imaging technologies can provide detailed characterization of crop attributes by capturing rich spectral information, supporting improved detection accuracy and operational decision-making in harvesting tasks such as maturity assessment, disease detection, and damage identification[103–108]. However, spectral imaging systems are associated with high equipment cost and data processing complexity, requiring trade-offs among detection accuracy, system cost, and processing efficiency in practical applications[109–112].

4.1.3. Near-Infrared Camera

Growing demand for real-time sensing of internal physiological states has promoted the use of near-infrared (NIR) imaging in crop harvesting. NIR imaging effectively detects key indicators such as moisture, sugar, and protein content due to strong tissue

penetration and high sensitivity to chemical composition. Unlike visible light imaging, which captures only surface color and texture, NIR imaging records how plant tissues absorb and reflect light in the 700–1400 nm wavelength range. Non-destructive detection of internal quality is thus enabled[113].

The working principle of near-infrared (NIR) imaging devices is based on the characteristic absorption behaviors of different substances in the NIR spectrum. For example, water exhibits a distinct absorption peak around 970 nm, while components such as sugars and starches show significant variations in the 1100–1400 nm wavelength range. Therefore, by capturing the reflectance changes of crops in these specific bands, it is possible to infer their moisture status, maturity level, and internal quality[114]. An example of such imaging is shown in Figure 15.

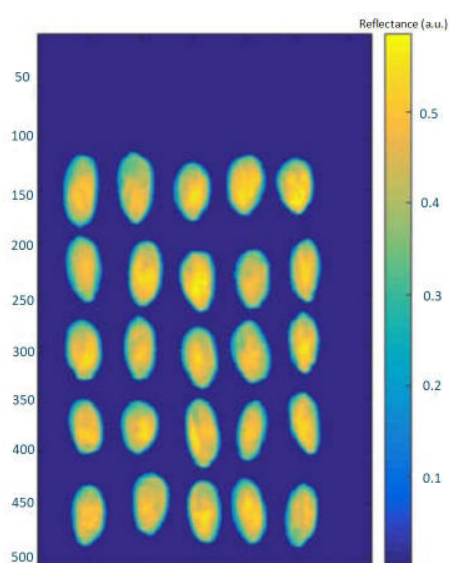


Figure 15. Goji berry hyperspectral data based on near-infrared camera. Reproduced from Kang et al.[115]; licensed under CC BY 4.0.

In practical applications, near-infrared (NIR) sensors can be embedded in operational platforms such as combine harvesters and fruit and vegetable picking robots to enable real-time monitoring of moisture content during the harvesting process. possible to accurately predict their dry matter accumulation level, thereby guiding the adjustment of threshing intensity and the zoned management of grain moisture. Harvesting parameters can thus be optimized, and breakage rates and moisture loss reduced. Similarly, in fruit picking processes, determining ripeness based on changes in NIR reflectance can improve picking accuracy[115]. In these scenarios, NIR sensing needs to maintain stable reflectance measurement under changing illumination and crop posture, while the detected moisture or maturity information should be transmitted in time to the onboard control system for threshing adjustment or picking decisions.

Traditional single-band near-infrared (NIR) detection offers advantages such as low cost and ease of deployment. However, it has limitations in terms of component differentiation accuracy and resistance to interference. To improve detection precision, recent developments have introduced multi-band NIR imaging and broad-spectrum inference methods. By simultaneously capturing reflectance information across multiple characteristic NIR wavelengths and applying modeling techniques such as Principal Component Analysis (PCA) and Partial Least Squares Regression (PLSR), it is possible to achieve rapid field-level detection of crop attributes under real-world agricultural conditions[116].

To meet the real-time detection demands of high-speed harvesting operations, near-infrared (NIR) imaging devices are increasingly being developed toward higher frame rates and wider dynamic ranges at the hardware level. At the same time, Edge computing and lightweight regression networks have been integrated into the NIR data processing pipeline to reduce transmission and inference latency. As a result, real-time feedback on crop status during harvester movement can be achieved by the detection system[117]. Rapid response of NIR imaging systems is especially valuable in applications such as monitoring grain moisture content, supporting intelligent adjustment of agricultural machinery parameters.

In summary, near-infrared (NIR) imaging enables non-destructive detection of crop moisture status, maturity, and damage by analyzing near-infrared reflectance, supporting harvest timing determination, quality grading, and anomaly screening[118]. NIR detection has advantages in acquiring internal quality information, with system costs between visible-light systems and high-end spectral equipment, showing certain engineering potential. However, it is sensitive to environmental conditions and often relies on indirect reflectance information, leading to uncertainty across crop types and growth stages; therefore, trade-offs among detection accuracy, system cost, and real-time performance are required in practical applications[119,120].

4.1.4. LiDAR

In agricultural harvesting, three-dimensional crop features such as plant height, canopy density, and spatial distribution are important for path planning and harvest timing. Traditional 2D imaging is limited by occlusion and scale variation, making it difficult to capture full spatial information. LiDAR (Light Detection and Ranging), an active laser-based sensor, can efficiently acquire accurate 3D point cloud data, making it widely used in modern intelligent agricultural machinery.

The working principle of LiDAR is based on emitting laser pulses and measuring their return time to precisely calculate the distance between the sensor and the target surface. By scanning the crop canopy, LiDAR reconstructs a high-precision 3D point cloud model of crop spatial distribution. Structural information such as plant height, canopy thickness, and lodging status can be extracted from this model, providing fundamental data support for optimizing harvesting strategies[121].

In field applications, LiDAR-based height measurements can effectively estimate crop growth progress and assist in determining the optimal harvesting window. For example, Tan developed a potato seedling canopy height detection platform based on phased-array LiDAR. By generating 3D point clouds through laser scanning, as shown in Figure 16, and combining the installation height (H_T), canopy height (H_p), ground reference height (H_L), scanning distance (D) and scanning spot (S_{ij}), canopy information can be accurately extracted[122].

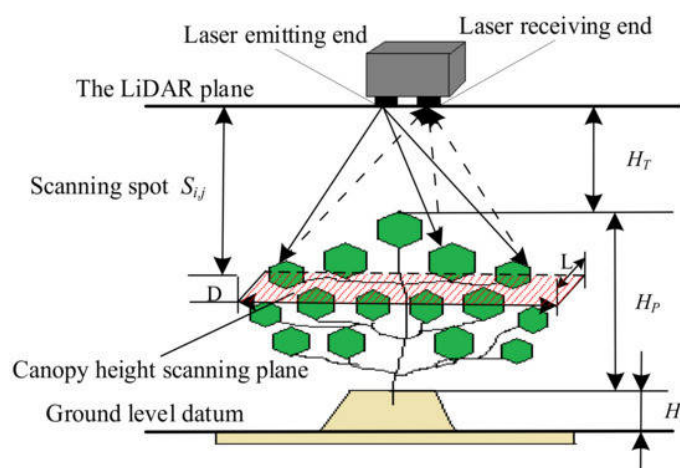


Figure 16. Potato seedling canopy height detection based on LiDAR. H_T : the installation height of the LiDAR; H_P : the canopy height; H_L : the ground reference height; D : the scanning distance; S_{ij} : the scanning spot. Reproduced from Tan et al.[122]; licensed under CC BY 4.0.

To address challenges such as uneven terrain and dense vegetation in field environments, modern agricultural LiDAR systems have gradually adopted multi-line laser scanning (e.g., 16-line and 32-line LiDAR) and adaptive point cloud filtering techniques to enhance spatial resolution and noise resistance[123]. By integrating Inertial Navigation Systems (INS) with Global Positioning Systems (GPS), laser point clouds can be spatially registered and linked to geographic coordinates, enabling large-scale crop monitoring and management[124].

As the core component of LiDAR applications, point cloud data processing has evolved from traditional geometric feature extraction to deep learning-based semantic segmentation. Traditional methods such as Random Sample Consensus (RANSAC) and terrain smoothing filters are mainly used for ground removal and vegetation separation, but they often produce errors in complex canopy structures. In recent years, point cloud direct processing models based on deep learning—such as PointNet and PointNet++ have been introduced into the agricultural domain. These models can classify and segment raw point clouds without relying on gridding or projection preprocessing, significantly improving the accuracy of 3D crop reconstruction and attribute extraction[125].

In harvesting applications, LiDAR systems can be embedded into platforms such as combine harvesters and autonomous tractors to dynamically perceive canopy density and height during operation. For example, real-time canopy analysis can determine crop maturity and guide header height adjustment, improving efficiency and reducing losses. In orchard operations, LiDAR combined with SLAM can assist autonomous navigation and precise target positioning for picking robots, improving harvesting success rates[126]. In such applications, LiDAR point clouds need to be processed and registered in time, so that canopy height, obstacle position, or row-structure information can be converted into header adjustment, navigation correction, or picking-path commands during machine movement. Therefore, maintaining robustness under vibration, dust, and sparse point-cloud conditions is critical for reliable system integration.

Despite its high precision and robustness in 3D perception, LiDAR still faces challenges such as high cost, complex data processing, and performance degradation under extreme conditions. To address these issues, current research focuses on low-cost solid-state LiDAR and point cloud sampling and compression techniques. In addition, RGB or near-infrared data fusion is explored to improve system cost-effectiveness and adaptability[127].

In summary, LiDAR provides high-precision 3D spatial data for agricultural harvesting, supporting crop height measurement, canopy structure modeling, path planning, and operational adjustment. Compared with 2D imaging methods, LiDAR has clear advantages in spatial structural representation; however, its high equipment cost, system complexity, and maintenance requirements limit its large-scale application in practical agricultural harvesting scenarios.

4.1.5. Other Sensors

In agricultural harvesting, besides mainstream optical, spectral, near-infrared cameras, and LiDAR, some less common sensors offer unique advantages in specific environments or for particular functions. These sensors are an important supplement to the construction of multi-source perception systems. Representative sensors include infrared thermography, millimeter-wave radar, and structured light cameras.

Infrared thermal imagers are passive sensors that form images by detecting heat emitted by objects, offering advantages such as independence from external light sources and usability at night. The surface temperature of crops varies during different growth stages or health conditions, and thermal imaging can reveal temperature changes related to water stress, pests, or the ripening process[128]. For example, the respiration of mature fruits and vegetables increases, causing localized heating, while diseased areas often show abnormal cold or hot spots. Non-contact and real-time imaging allow deployment on harvesting robots and machinery, enabling efficient online monitoring[129].

Millimeter-wave radar is an active sensor that measures distance and velocity based on millimeter-wave signal reflection. Its strong penetration and environmental robustness make it suitable for complex field conditions such as low light, dust, rain, and fog. During high-speed harvesting operations, it supports terrain recognition, obstacle detection, and crop boundary identification. For example, when a combine harvester turns or approaches field edges, it can detect large obstacles such as tree stumps or ridges, supporting path planning and obstacle avoidance. In addition, its high ranging accuracy and velocity detection enable dynamic monitoring of targets on fruit and vegetable conveyors, improving response speed and safety[130].

Structured light cameras are active 3D imaging devices that project coded light patterns and analyze their deformation to obtain depth information. Compared to LiDAR, they are suitable for mid- to short-range, high-density 3D modeling and provide higher resolution and data stability[131]. In fruit and vegetable harvesting, they are used for spatial separation of fruits and branches, and for reconstructing fruit contours and volume to support robotic path planning and gripper positioning[132]. They are relatively robust to occlusion and irregular shapes, reducing misjudgment during harvesting. Compared to stereo vision, they are less sensitive to lighting and do not require complex calibration, making them suitable for edge deployment on mobile platforms.

In summary, infrared thermal cameras, millimeter-wave radar, and structured light cameras offer distinct advantages in temperature, distance, and 3D shape perception, helping to complement the perceptual limitations of mainstream sensors and expand the sensing dimensions of agricultural harvesting systems. However, their practical application is still constrained by environmental sensitivity, spatial resolution, measurement range, and system integration complexity, while multi-source heterogeneous data alignment and real-time processing remain challenging.

4.2. Datasets for Crop Attribute Detection

In recent years, with the development of crop attribute detection tasks, relevant datasets have been gradually constructed to support the methods discussed in this paper, including image classification, object detection, image segmentation, and point cloud

analysis. In image classification-based crop attribute detection, some datasets for crop type recognition and quality evaluation have been developed, such as PlantVillage and Fruit-360, which can be used to support classification tasks related to maturity, disease, and quality attributes. In object detection-based crop attribute detection, datasets such as the Global Wheat Head Detection (GWHD), Maize Tassel Dataset, and Apple Detection Dataset have been developed, which provide bounding box annotations for tasks such as spike detection and crop density estimation and can be used to train models such as YOLO and Faster R-CNN. In image segmentation-based crop attribute detection, datasets such as VegAnn, CropWeed Field Image Dataset, and Rice Segmentation Dataset have been developed, which provide pixel-level annotation information for tasks such as crop region extraction and lodging area segmentation, thereby supporting canopy structure analysis and lodging-related attribute detection. In point cloud-based crop attribute detection, datasets such as Crops3D and Pheno4D have been developed, which are used to describe crop spatial structures and three-dimensional morphological characteristics, providing data support for lodging attribute detection and spatial distribution analysis. In other attribute detection methods, such as tensor decomposition and graph neural networks, multi-source heterogeneous data are usually used to construct analytical datasets, such as the multi-temporal and multi-modal data provided by agricultural data platforms like Radiant Earth MLHub and OpenSAR, which can be used for modeling crop growth processes and analyzing and predicting attributes such as yield and structural characteristics. Overall, these datasets have been constructed around key tasks in crop attribute detection and have provided data support for crop attribute detection methods based on image classification, object detection, image segmentation, and point cloud analysis.

Although the above datasets have supported the development of crop attribute detection methods to a certain extent, many limitations still exist from the perspective of harvesting applications. First, most datasets are limited in scale and are usually constructed for a single crop or specific environment, making it difficult to meet the requirements of complex field conditions and multi-crop detection scenarios addressed in this paper; second, different datasets lack unified standards in terms of data acquisition methods, annotation standards, and evaluation metrics, which makes it difficult to directly compare the performance of different crop attribute detection methods; finally, in actual harvesting operations, factors such as occlusion, dust, and dynamic environments are widely present, while datasets constructed specifically for such complex conditions are still relatively scarce. These issues limit the generalization ability and robustness of the methods reviewed in this paper in practical applications. Therefore, constructing large-scale, diverse, and standardized datasets oriented toward harvesting scenarios is a key factor in promoting the practical application of crop attribute detection technologies.

4.3. Approaches of Data Analysis and Crop Attribute Detection

In modern agricultural harvesting, relying solely on sensor data acquisition is no longer sufficient to meet the requirements for accuracy, real-time performance, and adaptability. Therefore, data analysis and feature extraction methods are essential for transforming raw sensor data into usable crop attribute information. Existing methods can mainly be classified into image segmentation, image classification, object detection, and point cloud analysis (Figure 17). For harvesting machinery, these methods should not only provide accurate recognition or estimation results, but also generate information in time for operational decisions, such as target selection, path adjustment, header control, and end-effector motion. Therefore, this section analyzes the applicability of different methods in crop attribute detection from the perspectives of feature representation capability, computational efficiency, robustness under complex harvesting conditions, and their connection with machinery decision-making and control.

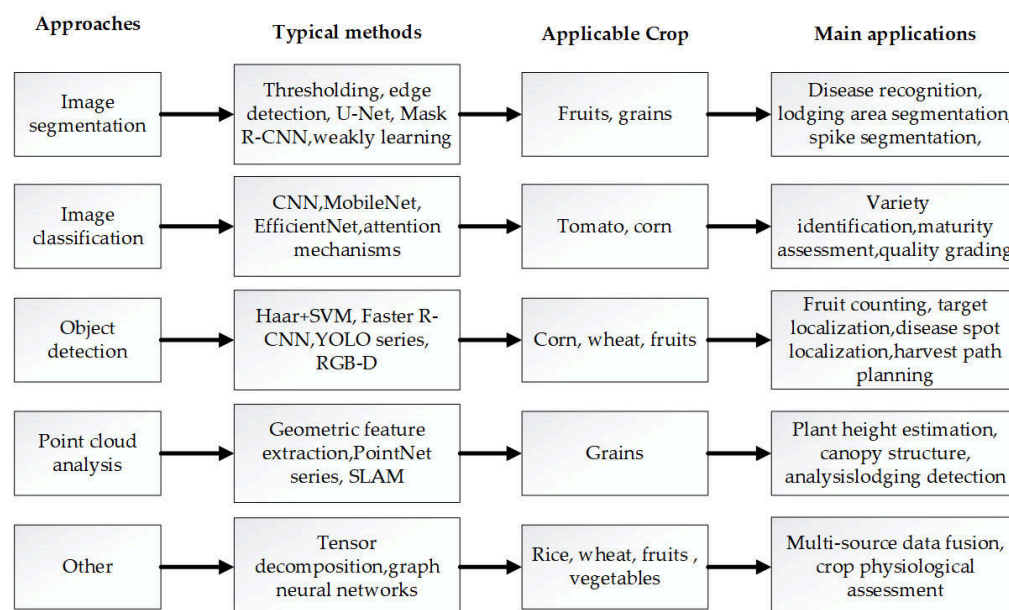


Figure 17. Overview of data analysis and crop attribute detection approaches.

4.3.1. Image Segmentation

Image segmentation is used to partition agricultural images into meaningful regions and to delineate crop-related areas at the pixel level. Through this process, detailed spatial features, including lesion regions, leaf edges, lodging boundaries, and harvestable areas, can be obtained from complex field backgrounds. Unlike object detection or attribute regression, which mainly provide bounding boxes or numerical estimates, segmentation produces fine-grained masks that preserve the shape and boundary information of target regions. Therefore, it has been widely applied to tasks such as crop disease analysis, maturity-region delineation, spike or ear contour extraction, and labeling of harvestable regions[13].

Traditional segmentation in agricultural images was usually implemented through intensity- or color-based threshold selection, together with edge detection operators. These methods identify crop-related regions by separating pixels according to predefined or automatically determined grayscale and color criteria. Representative approaches include Otsu thresholding and Canny edge detection, which can produce acceptable results when illumination and background conditions are relatively stable[14]. However, their performance often deteriorates in field scenes because variable light, cast shadows, and complex backgrounds can change pixel distributions and weaken object boundaries. To improve segmentation robustness, later studies incorporated color-space conversion, such as HSV and Lab representations[15], together with morphological post-processing, including erosion, dilation, and closing, to refine segmented regions and reduce noise.

With the increasing availability of pixel-level annotations, supervised semantic segmentation networks have become important tools for crop image analysis. FCN introduced a fully convolutional prediction framework that allows dense category assignment for each image pixel, providing an early basis for deep-learning-based agricultural segmentation. On this foundation, U-Net further improved feature recovery by combining encoder-derived semantic information with decoder-stage spatial details through skip connections. This structure has therefore been widely adopted in tasks such as leaf delineation and lesion-area extraction[16]. In applications involving wheat rust and apple black spot, U-Net-derived models have often achieved segmentation accuracies above 90%.

Instance segmentation extends semantic segmentation by distinguishing different object instances while also generating their corresponding masks. This capability is useful

in dense crop scenes where neighboring plants, fruits, or ears may overlap. Mask R-CNN is a representative method of this type and has been applied to tasks such as fruit localization and ear separation. In corn ear segmentation, for example, its region proposal followed by mask prediction strategy can separate adjacent ears more effectively, thereby supporting subsequent counting and robotic path planning[17]. However, the region-proposal-based two-stage design also increases computational burden and inference latency, which makes real-time deployment on embedded harvesting platforms more difficult.

Based on typical convolutional neural network methods such as U-Net and Mask R-CNN, segmentation models have been further developed in terms of architecture design and feature modeling. For example, the DeepLab series (e.g., DeepLabv3+) enhances multi-scale feature representation by introducing atrous convolution and atrous spatial pyramid pooling (ASPP), showing good adaptability in scenarios with significant crop scale variation. When applied to harvesting machinery, such segmentation results need to be generated before the machine reaches the detected lodging or harvestable region, so that path correction or header adjustment can be performed in time. However, these methods still mainly rely on local convolutional features, which may be affected by missing local information in cases of severe occlusion.

In recent years, Transformer-based segmentation models have become an important complement to traditional convolutional approaches. Unlike convolution-based methods, Transformers establish global feature relationships through self-attention mechanisms, enabling better modeling of long-range dependencies and improved performance in occluded scenarios. For example, models such as Segformer and Mask2Former have demonstrated strong segmentation performance in complex environments. However, these models generally involve higher computational complexity, and their deployment in real-time agricultural harvesting scenarios requires balancing segmentation accuracy and computational efficiency.

For disease spot detection, image segmentation combined with attention mechanisms (e.g., SE or CBAM in U-Net) improves recognition of small targets and irregular boundaries under complex conditions. To reduce annotation costs, weakly and semi-supervised segmentation methods use image-level labels, point annotations, or pseudo-labels. This reduces annotation burden and improves practicality for large-scale disease surveys[18].

In harvesting, image segmentation is used for disease detection, lodging segmentation, ear contour extraction, and fruit integrity assessment. For example, lodging segmentation supports statistical analysis and path adjustment for autonomous machines. In fruit picking, instance segmentation separates occluded fruits from background structures, improving robotic positioning and stability[19].

In summary, image segmentation has evolved from rule-based methods to high-precision perception approaches integrating deep semantic modeling, attention mechanisms, and weakly supervised learning, significantly improving the spatial resolution capability of intelligent harvesting systems. By performing pixel-level delineation of crop organs, lodging areas, disease regions, or fruit boundaries, image segmentation methods can support local operation decisions such as cutting area identification, target separation, and quality screening. However, these methods still depend on high-quality annotations, are sensitive to complex backgrounds and illumination variation, and may degrade when detecting small or irregular targets. Meanwhile, high-resolution segmentation models often involve high computational cost, limiting their real-time application in harvesting systems.

4.3.2. Image Classification

Image classification methods are widely used in agricultural harvesting attribute detection, particularly for tasks such as variety identification, maturity assessment, and quality grading. The core of image classification is to map the entire image to a specific

category, enabling the identification of crop status. Rapid and accurate assessment of crop attributes during or prior to harvesting supports intelligent grading, timely picking, and precise task scheduling. Such assessment serves as a crucial component in the agricultural information perception process[6].

Before deep learning was widely adopted, crop image classification mainly depended on handcrafted visual descriptors, including color distribution, texture patterns, and contour-related features. Such descriptors were useful for simple tasks such as crop or fruit category recognition and preliminary health evaluation when images were collected under stable illumination and relatively uncluttered backgrounds. However, their reliability decreases markedly in field scenes, where illumination fluctuation, overlapping plants, and occluded fruits can alter the extracted visual features and lead to classification errors. Moreover, the design of these descriptors usually requires expert experience and crop-specific prior knowledge, which limits their transferability across crops, environments, and large datasets with ambiguous class boundaries[20].

To overcome the aforementioned limitations, Convolutional Neural Networks (CNNs) are widely used in agricultural image classification, extracting hierarchical features to improve accuracy and generalization[21]. In harvesting, CNN-based systems distinguish crop varieties and maturity stages to support harvesting strategies and parameter adjustment.

To meet real-time requirements, lightweight models such as MobileNet and EfficientNet reduce parameters through techniques like depth-wise separable convolution, enabling deployment on embedded devices for tasks such as maturity classification and crop grading. These models can achieve low-latency inference in field environments, meeting real-time requirements of harvesting equipment[22]. In robotic harvesting operations, such low-latency classification is important because the maturity or grade category needs to be determined before the robot executes target picking, sorting, or rejection actions.

Attention mechanisms embedded in CNNs enhance regional feature perception and improve robustness under background interference and variable fruit positions. By integrating prior knowledge and temporal information, image classification supports fruit grading, harvest timing, and variety identification.

In summary, image classification methods have advantages such as flexible deployment and clear model structure, and are suitable for scenarios with distinct global features and relatively stable class distributions. However, due to the lack of spatial location information, they remain limited in multi-target localization and analysis tasks. Meanwhile, classification models still require a trade-off between accuracy improvement and computational efficiency, and are therefore often combined with methods such as object detection to serve auxiliary functions such as preliminary screening or attribute recognition in multi-stage perception systems, provide basic decision-making information for maturity judgment, quality grading, and harvestability assessment.

4.3.3. Object Detection

Unlike image classification, which assigns a global label to an input image, object detection provides both semantic and geometric information for individual targets. It can identify the target class and estimate its location and size within the image, making it suitable for multi-object perception in harvesting scenes. Accordingly, object detection is widely used for fruit counting, ear or spike tracking, lesion localization, and harvest-route planning. In systems such as combine harvesters and robotic picking platforms, this capability supports field target recognition and real-time localization under dynamic operating conditions.

Before deep-learning-based detectors became dominant, object detection in agricultural images was mainly implemented through window scanning and handcrafted

feature-based recognition. Typical approaches used descriptors such as Haar-like features together with classifiers such as SVM to identify fruits, pests, or diseased regions. Although these methods provided early solutions for target localization, their feature representation ability was limited, and repeated region scanning usually resulted in high computational cost. As a result, they were difficult to adapt to harvesting machinery applications that require both fast response and reliable detection accuracy[23].

Deep-learning-based detectors have greatly improved the ability of agricultural vision systems to identify and localize crop targets. Faster R-CNN, as a representative two-stage detector, first generates candidate regions through a Region Proposal Network and then performs classification and bounding-box refinement, which enables reliable detection of small, dense, or irregular targets. In harvesting tasks involving tomatoes or grapes, it can help separate fruits from leaves, determine target positions, and provide visual information for robotic arm trajectory planning and stem-cutting operations[24]. However, its proposal generation and refinement stages increase computational cost and inference time, limiting its suitability for embedded harvesting platforms with strict real-time requirements. In contrast, one-stage detectors represented by the YOLO family perform target classification and bounding-box regression within a unified framework, reducing inference latency and improving processing efficiency. Therefore, YOLOv5, YOLOv7, and related variants have been widely used in fruit recognition and counting on high-speed harvesting platforms, providing a practical balance between detection accuracy and processing speed[25]. Compared with two-stage detectors, YOLO-type models generally require fewer computational steps and are more suitable for embedded harvesting devices, although their performance may decline under small-target, dense, or heavily occluded conditions[26–28]. As localization tasks increasingly require depth and structural information, object detection has also been extended from two-dimensional RGB images to RGB-D fusion and point-cloud-based perception[29]. Depth cameras, LiDAR, and structured-light sensors provide geometric cues in addition to color and texture features, supporting more accurate three-dimensional target localization in tasks such as maize ear positioning and fruit coordinate extraction[30]. For example, RGB-D-based detection can improve the separation of overlapping fruits and reduce errors caused by scale variation, thereby enhancing the stability of multi-target detection in complex harvesting scenes[31]. When the detected fruit positions can be mapped to the robot coordinate system in time, the results can be directly used for grasping-point selection and collision avoidance during end-effector motion. However, this process requires robust depth estimation under occlusion and reliable synchronization between visual perception and robot control.

In harvesting scenarios with varying lighting, complex poses, and motion blur, target detection models require strong generalization and real-time performance, achieved through strategies such as data augmentation, attention mechanisms, and multi-scale feature fusion[32].

In summary, object detection methods provide important support for real-time perception in crop harvesting and have evolved from traditional approaches to deep learning-based techniques for multi-target localization, fruit tracking, and spatial coordinate extraction. The detected targets and their spatial position information can provide direct information for fruit picking, spike counting, obstacle avoidance, and actuator path planning. However, in complex harvesting environments, occlusion, scale variation, and motion blur still affect detection accuracy and stability. Meanwhile, one-stage methods offer higher efficiency but may be limited in dense or small-target scenarios, whereas two-stage methods provide higher accuracy but involve greater computational cost, restricting real-time application.

4.3.4. Point Cloud Analysis

Point cloud analysis represents crops as sets of spatially distributed three-dimensional points and is used to describe their geometric morphology and field distribution. Each point records its position in 3D space through X, Y, and Z coordinates and may also carry additional information, such as intensity, reflectance, or color, depending on the sensor type. Such data are commonly acquired using LiDAR, structured-light cameras, or RGB-D sensors. Unlike conventional 2D image analysis, point cloud analysis preserves depth and spatial relationships, making it more suitable for reconstructing crop structures and describing attributes such as plant height, canopy architecture, lodging form, population density, and crop location[33].

Point cloud technology in agriculture has evolved from static 3D reconstruction and height estimation to geometric feature extraction and learning-based analysis. Traditional methods apply spatial filtering, normal estimation, and curvature analysis to build crop models and extract features such as canopy hierarchy, porosity, and vertical variation, supporting path optimization and parameter adjustment; lodging detection further uses surface normal variation and tilt trends to evaluate crop uprightness for machine control. Recent deep learning approaches, including PointNet, PointNet++, and PointConv, directly model raw point clouds for classification, segmentation, and key point extraction, enabling applications such as branch and leaf recognition, corn ear counting, and fruit segmentation. For example, a 3D phenotyping system for apple trees achieved an R^2 of 0.987 in height prediction and 93.4% accuracy in branch counting[34]. For robotic harvesting, such structural information can support manipulator workspace judgment and picking-trajectory planning. However, when branch overlap, leaf occlusion, and viewpoint changes cause incomplete point-cloud reconstruction, the robustness of the analysis results still needs to be ensured.

To enhance operational reliability and facilitate field deployment, recent studies have combined image data with point cloud information in crop perception systems. In such frameworks, RGB images contribute appearance cues, including color and surface texture, whereas point clouds provide geometric information related to depth, shape, and spatial arrangement. The complementary use of these two data types enables a more complete representation of crop targets and improves perception robustness under occlusion, background clutter, and sensor interference in field environments. Moreover, SLAM has been incorporated into harvesting perception platforms to support simultaneous environment mapping and machine localization. When coupled with real-time point cloud processing, SLAM-based perception can assist agricultural machinery in maintaining navigation accuracy and tracking crop targets continuously during field operation[35].

In summary, point cloud analysis methods have evolved from simple geometric fitting to 3D perception frameworks integrating deep learning, spatiotemporal modeling, and multi-modal fusion. Their high spatial resolution and ability to represent complex structures make them valuable for harvesting regulation tasks that rely on 3D information. The extracted 3D structural parameters, such as plant height, canopy geometry, lodging angle, and target position, can support header adjustment, navigation correction, picking trajectory planning, and operation parameter optimization. However, these methods are sensitive to data quality, as point cloud noise, sparsity, and missing data can affect structural reconstruction and feature extraction. Meanwhile, large-scale point cloud processing involves high computational cost, and multi-modal fusion requires accurate calibration and registration, increasing system complexity.

4.3.5. Other Methods

In agricultural harvesting, mainstream technologies include image classification, object detection, image segmentation, and point cloud analysis. In addition, methods such

as density regression and multimodal fusion have also been used to address less common issues, including special target recognition, limited samples, and specific interference conditions[20]. These methods can provide auxiliary data information for harvesting decision-making and intelligent operation control.

Tensor decomposition and multidimensional modeling provide effective tools for analyzing multi-source heterogeneous agricultural data with spatiotemporal characteristics. By representing data such as remote sensing images, hyperspectral maps, and multi-sensor observations as tensors and applying methods like CP or Tucker decomposition, key features and evolution patterns can be extracted, supporting tasks such as crop growth monitoring and optimal harvest timing[36]. In addition, Graph Neural Networks (GNNs) model complex spatial relationships in farmland by representing objects as nodes and their interactions as edges, enabling analysis of non-Euclidean data. These methods have been applied to tasks such as yield prediction, 3D phenotyping, and agroecosystem modeling. For example, a GNN-based model integrating multi-source data achieved an R^2 of 0.70 and a prediction error of 0.17 t/ha for winter wheat yield, representing a 28.6% improvement over previous methods[37].

Although these emerging methods have not yet been widely applied in agricultural harvesting perception systems, they have shown advantages in adapting to complex environments and integrating multi-source data, providing complementary support to traditional methods through cross-modal and high-dimensional feature representation. However, their application is still constrained by insufficient standardized datasets, high model complexity, and difficulties in stable integration with existing perception pipelines. Meanwhile, field variability and data heterogeneity may also affect their generalization and robustness.

5. Challenges in Crop Attribute Detection

Despite significant advances in sensing, data processing, and model recognition, intelligent agricultural perception systems still face many challenges in practical harvesting operations. Dynamic field environments, uncertain crop states, and high-speed machinery operations place higher demands on the stability, robustness, and real-time performance of perception systems. These systems must accurately detect crop position, maturity, and quality under complex conditions such as variable illumination, weather, terrain, and occlusion, while operating with limited computational resources. This chapter systematically analyzes the above key challenges, examines their underlying mechanisms from the perspectives of environmental variation, computational constraints, and dynamic interference, and discusses their impacts on the accuracy, stability, and real-time performance of crop attribute detection.

5.1. Complex Harvesting Environment

In agricultural production, the highly variable conditions—such as lighting, weather, wind speed, humidity, and crop growth states—can significantly impact detection accuracy. Detection systems must reliably perform attribute recognition, disease identification, and yield estimation under diverse environmental conditions and background interference[133]. The main challenges include lighting disturbances, weather effects, leaf occlusion, and background noise, as shown in Figure 18. Techniques such as illumination compensation, data augmentation[134], sensor fusion, and anti-interference algorithms can improve environmental adaptability. The application of these technologies is expected to improve perception stability under complex field conditions and provide a more reliable information basis for mechanical control.

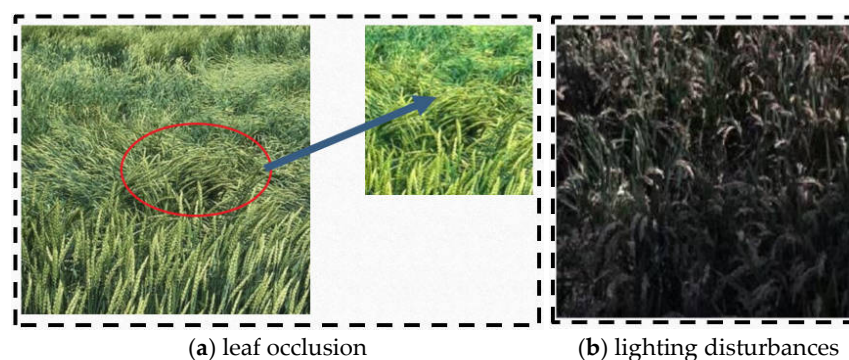


Figure 18. Examples of complex harvesting environments.

In natural environments, lighting conditions vary with time, weather, and sun angle, significantly affecting crop image acquisition. Weak lighting can cause insufficient brightness and unstable features, while strong sunlight may lead to overexposure and detail loss; cloudy conditions reduce contrast, and occlusion or reflections introduce shadows and glare, increasing detection difficulty and reducing accuracy in maturity and disease analysis. To mitigate these effects, adaptive exposure control and automatic white balance can improve image consistency under varying conditions. Image enhancement methods such as histogram equalization, gamma correction, and Retinex can further improve contrast and reduce illumination interference[135]. In addition, multi-source lighting (e.g., LEDs or near-infrared) can enhance detection in low-light environments[136], while data augmentation and illumination normalization improve model generalization during deep learning training.

To address crop occlusion, deep learning-based semantic segmentation can separate leaves and fruits to improve detection accuracy. For background interference, methods such as GrabCut and background modeling remove non-target areas like weeds and soil. In addition, hyperspectral imaging uses spectral features to distinguish crops from the background, improving classification accuracy[137]. These methods are applied in harvesting path optimization, precise localization, and fruit feature extraction.

To address weather disturbances such as wind, rainfall, and humidity, optical flow and LSTM-based methods predict crop motion and perform compensation to improve detection accuracy. UAVs and agricultural robots can integrate gyroscopes and inertial navigation systems to enhance image stability. In rainy or high-humidity environments, image enhancement methods such as dark channel prior and Retinex improve image quality[138]. In addition, high-protection sensors and multi-sensor fusion (RGB, infrared, and LiDAR) mitigate weather interference and improve system robustness[139].

Practical harvesting scenarios often involve multiple interference factors, such as low illumination, dust, rainfall, strong reflections, plant occlusion, and mechanical vibration. These factors may cause image blurring, contrast degradation, reduced signal-to-noise ratio, feature distortion, and unstable sensor data, thereby increasing the difficulty of stable feature extraction and accurate recognition. Existing studies have proposed corresponding improvements for the above individual or multiple types of problems. However, high-precision real-time detection in complex harvesting operations under combined interference factors has not yet been effectively addressed and requires further investigation.

In summary, complex field environments pose multiple challenges to crop detection technologies, including illumination variation, weather conditions, crop occlusion, and background interference. These factors not only affect the stability of feature extraction but also introduce uncertainty in model inference under dynamic and unstructured harvesting scenarios. Although image preprocessing, sensor fusion, deep learning models, and intelligent compensation algorithms can improve system adaptability, their

effectiveness often depends on data quality and scenario-specific adjustment, limiting cross-environment generalization.

5.2. High Real-Time Requirement

In agricultural harvesting, real-time detection is critical to crop quality and operational efficiency. Machinery must rapidly detect indicators such as grain loss, impurities, and ear number and adjust parameters accordingly. For example, a combine harvester measures crop moisture as it enters the header to guide threshing and cleaning adjustments, reducing loss and improving quality, as shown in Figure 19[140]. Detection delays may cause over-threshing, grain breakage, or reduced efficiency, requiring high system response speed.

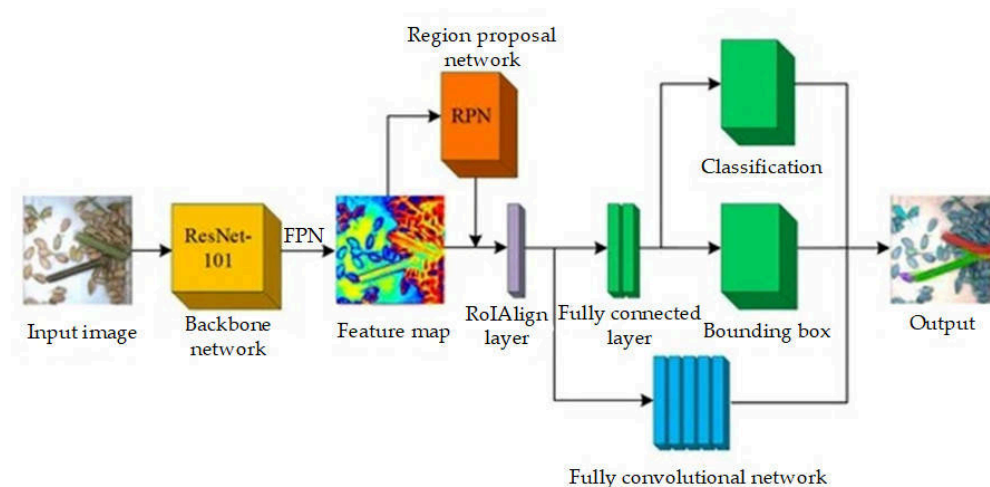


Figure 19. Flowchart of rice impurity recognition algorithm based on Mask R-CNN. Reproduced from Guan et al.[140]; licensed under CC BY 4.0.

To meet the real-time detection requirements of high-speed harvesting, modern detection systems commonly employ high-frame-rate industrial cameras, near-infrared sensors, GPU parallel computing, and edge computing. High-frame-rate cameras can capture color and morphological features within milliseconds, while near-infrared sensors rapidly analyze moisture content to determine crop maturity in real time. GPUs and edge computing enable immediate on-machine data processing, reducing latency and enhancing response speed[141]. Especially in large-scale harvesting operations, real-time processing capability is key to achieving rapid feedback and precise adjustment.

Although edge computing can reduce data transmission latency and improve real-time performance in harvesting scenarios, its practical application is still constrained by the limited power supply capacity and computational resources of onboard embedded systems. Therefore, it is necessary to achieve a balance between detection performance and system resource consumption through lightweight model design and energy-efficient inference methods.

In addition to hardware improvements, algorithm-level optimization plays a key role in enhancing detection speed. Lightweight models such as MobileNet and YOLOv5-Nano are widely used in agricultural scenarios, enabling efficient target detection and attribute recognition on edge devices. Model pruning and quantization further reduce inference time by removing redundant parameters and simplifying computations while maintaining accuracy. For example, a YOLOv5 model optimized with these techniques achieved over a 40% increase in inference speed in fruit detection tasks, supporting real-time picking and sorting applications[142].

In summary, detection response speed is a key factor in agricultural harvesting automation. Delays can disrupt the synchronization between crop perception and parameter adjustment during high-speed operation, affecting operational quality and system stability. Lightweight models, model compression, and edge deployment can improve response performance, but usually require trade-offs between computational efficiency and detection accuracy. Meanwhile, hardware platform differences and communication latency may also limit practical system responsiveness.

5.3. Motion Interference of Agricultural Harvesting Machinery

During agricultural harvesting, high-speed operation and uneven terrain cause vibrations and motion disturbances that interfere with sensor data collection. These effects degrade image processing performance and reduce detection accuracy and stability. The impact is particularly significant for sensors such as cameras, LiDAR, and infrared imaging systems.

The visual blur caused by the motion of agricultural machinery has a significant impact on detection performance. Due to the posture changes of the camera caused by machine vibrations, phenomena such as misalignment of crop contours, edge deformation, and target occlusion can occur in the images. In severe cases, motion blur may arise, significantly reducing image clarity and algorithm recognition accuracy, as shown in Figure 20. This is particularly problematic in tasks that rely heavily on small target recognition or boundary details, as image blurring may lead to feature extraction failure and increased detection errors[143].



Figure 20. Examples of motion interference in agricultural harvesting machinery.

When agricultural machinery operates at varying speeds, turns, or moves on slopes, frequent changes in posture (pitch, roll, and yaw) can cause point cloud distortion and image deformation, reducing the accuracy of spatial reconstruction and object detection. In most existing studies on crop attribute detection, calibration is performed under static conditions, while real-time compensation for posture changes and motion disturbances remains insufficient, which can reduce system adaptability in complex harvesting environments. Therefore, some researchers have begun to adopt methods such as image deblurring, motion compensation, and multi-sensor calibration for improvement. For example, CNN-based deblurring models can improve image quality to some extent under motion blur, while visual-inertial systems combining IMUs and cameras can also be used for dynamic posture compensation to improve system stability and positioning accuracy[144]. In addition, multi-view acquisition and image fusion can provide complementary information when single-frame data are unreliable. However, under high-speed operation, severe vibration, and complex terrain conditions, real-time compensation accuracy, cross-sensor synchronization, and model inference efficiency still face limitations. Therefore, stable perception and real-time compensation under motion interference still need to be further strengthened.

In summary, machinery motion remains a key bottleneck affecting crop detection accuracy in automated field operations. Vibration, motion blur, and viewpoint changes can interfere with feature extraction and perception consistency. Improved sensor robustness, image stabilization, and multi-source fusion can partially mitigate these problems, but their effectiveness is still constrained by dynamic operating environments and multi-sensor synchronization requirements. In addition, motion-induced disturbances may accumulate during continuous operations, further affecting detection stability.

6. Future Development Trends

With the advancement of agricultural intelligence, crop attribute detection is evolving from single sensing to multi-dimensional integration and dynamic decision-making. Despite progress in sensing and algorithms, challenges remain in generalization, collaboration, and environmental adaptability. Future work should focus on flexible perception frameworks and improvements in learning mechanisms, system architectures, and resource scheduling. At the same time, the application of artificial intelligence in agricultural decision-making introduces challenges related to reliability and interpretability. Ensuring the trustworthiness of decision-making results is therefore a key prerequisite for practical deployment. This chapter focuses on future directions such as weakly supervised learning, few-shot learning, and multi-agent collaborative architectures, with emphasis on their roles in improving data utilization efficiency, enhancing system generalization, and supporting collaborative perception and decision-making.

6.1. Weak Supervision and Few-Shot Learning

In agricultural harvesting, crop attribute detection is challenged by limited annotated data, high labeling costs, and ambiguous class differences. Complex field conditions, including seasonal variation, crop diversity, and weather changes, further hinder the collection of large-scale, high-quality datasets. Traditional supervised deep learning methods often suffer from overfitting and performance degradation under data scarcity. Weak supervision and few-shot learning therefore represent key directions for improving agricultural perception systems.

Weakly supervised learning trains models using incomplete or imprecise labels. In agricultural detection tasks, image-level tags, bounding boxes, or sparse point annotations can replace dense pixel-level labels. Methods such as multiple instance learning, pseudo-labeling, and self-supervised learning improve model generalization. In addition, spatial priors from remote sensing data and historical operational data can serve as auxiliary supervision, reducing reliance on manual annotation[145].

Few-shot learning enables rapid generalization with limited samples and is suitable for scenarios such as new crop identification, rare condition recognition, and novel disease detection. Meta-learning methods such as MAML and Prototypical Networks optimize models through few-shot task simulation, enabling fast convergence with minimal data[146]. Incorporating domain-specific priors, such as agronomic knowledge or crop growth models, further enhances performance. Data augmentation also plays a key role, with techniques such as GAN-based image synthesis and geometric or illumination transformations improving robustness in real-world deployment[147].

In summary, weakly supervised and few-shot learning can alleviate data scarcity and improve the adaptability and generalization of agricultural vision systems, making them suitable for rapid deployment on new crops, emergency detection in extreme scenarios, and cross-regional model transfer. However, these methods usually rely on limited or noisy annotations, which may affect model stability and accuracy in complex field environments. Meanwhile, few-shot methods have limited adaptability to intra-class

variations under different growth stages and environmental conditions, while weakly supervised methods may be affected by label uncertainty in fine-grained recognition.

6.2. Multi-Agent Collaborative Perception System

As agricultural operations expand and tasks become more complex, single-platform detection systems can no longer meet large-scale, heterogeneous perception demands. A key direction is the development of multi-agent collaborative systems involving drones, agricultural machinery, ground robots, and edge nodes. Through coordinated sensing and information sharing, these systems enable large-scale, high-precision, and robust crop perception and harvest control.

The key advantage of multi-agent systems lies in spatial and functional complementarity. Aerial drones provide large-scale, high-mobility global observation for mapping crop distribution and detecting lodging, while ground machinery and robots offer higher local resolution for tasks such as fruit recognition and quality assessment. Their coordination enables a macro–micro perception framework that improves efficiency and adaptability. Effective collaboration requires distributed perception and decision-making, allowing agents to operate autonomously while sharing information for global coordination. In addition, reinforcement learning and graph neural networks can optimize task allocation and coordination, improving resource utilization and system efficiency.

In summary, multi-agent systems are suitable for complex agricultural environments with limited single-platform coverage, such as orchards, mixed-crop fields, and mountainous farmlands. With advances in 5G, satellite–terrestrial communication, and robotic platforms, they can enable collaborative perception and large-scale deployment, supporting continuous crop monitoring and automated operations. However, communication latency, packet loss, asynchronous data acquisition, and time-alignment errors may affect real-time coordination and perception consistency. Meanwhile, differences in sensing capabilities among platforms increase the difficulty of data fusion and task allocation, while larger system scales place higher demands on bandwidth and computational resources.

7. Conclusions

This review addresses crop attribute detection in the context of intelligent agricultural harvesting machinery. This review focuses on the attributes of grain, fruit, and vegetable crops from the perspectives of harvesting efficiency, loss rate, operating speed, and failure rate. It also analyzes the sensors and algorithms involved in detecting these attributes and clarifies the relationship between detection methods and control decisions in agricultural harvesting machinery. Key attributes such as plant height, canopy structure, density, spike number, spatial positioning, maturity, and quality are discussed, along with their roles in operational decision-making. This review further aims to systematically summarize crop attribute detection requirements, analyze key sensing technologies and data analysis methods, and identify current challenges and future development directions in intelligent harvesting systems. The application characteristics of mainstream sensors and the evolution of core methods, including image classification, object detection, semantic segmentation, and point cloud analysis, are summarized.

Recent advances in sensing and data analysis have improved detection accuracy and promoted more precise and automated harvesting. However, crop detection in real harvesting environments still faces challenges from complex field conditions, real-time constraints, and motion interference. Factors such as illumination variation, occlusion, uneven terrain, and high-speed operation increase sensor noise and reduce system reliability.

Future research should focus on improving multi-sensor fusion adaptability, developing lightweight and efficient models for real-time deployment, and integrating crop detection with agricultural big data for intelligent decision-making. Crop detection is

evolving toward multimodal perception, multi-task modeling, and system-level intelligent control. These developments are expected to support more efficient and precise harvesting operations across a wider range of application scenarios and further promote the improvement of intelligent control during the harvesting process.

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Abbreviations

The following abbreviations are used in this manuscript:

RGB	Red, Green, Blue
LiDAR	Light Detection and Ranging
UAV	Unmanned Aerial Vehicle
CNN	Convolutional Neural Network
R-CNN	Region-Based Convolutional Neural Network
YOLO	You Only Look Once
FCN	Fully Convolutional Network
GNSS	Global Navigation Satellite System
IMU	Inertial Measurement Unit
SLAM	Simultaneous Localization and Mapping
NIR	Near Infrared
HSI	Hyperspectral Imaging
NDVI	Normalized Difference Vegetation Index
GLCM	Gray Level Co-occurrence Matrix
SVM	Support Vector Machine
RF	Random Forest
PCA	Principal Component Analysis
PLSR	Partial Least Squares Regression
CHM	Crop Height Map
DSM	Digital Surface Model
DTM	Digital Terrain Model

References

1. Zhang, C.; Kovacs, J.M. The application of small unmanned aerial systems for precision agriculture: A review. *Precis. Agric.* **2012**, *13*, 693–712. <https://doi.org/10.1007/s11119-012-9274-5>.
2. Liakos, K.G.; Busato, P.; Moshou, D.; Pearson, S.; Bochtis, D. Machine learning in agriculture: A review. *Sensors* **2018**, *18*, 2674. <https://doi.org/10.3390/s18082674>.
3. Feng, Y.; Liao, D.; Wu, X.; Gao, R.; Fan, Y.; Raza, M.A.; Wang, X.; Yong, T.; Liu, W.; Liu, J.; et al. Effect of aboveground and belowground interactions on the intercrop yields in maize–soybean relay intercropping systems. *Field Crops Res.* **2017**, *203*, 16–23. <https://doi.org/10.1016/j.fcr.2016.12.007>.
4. GB/T 35381.1–2017; Tractors and machinery for agriculture and forestry—Serial control and communications data network—Part 1: General standard for mobile data communications. Standards Press of China: Beijing, China, 2017. Available online: <https://std.samr.gov.cn/gb/search/gbDetailed?id=71F772D82730D3A7E05397BE0A0AB82A> (accessed on 13 March 2026).
5. Ji, Y.; Mestrot, A.; Schulin, R.; Tandy, S. Uptake and transformation of methylated and inorganic antimony in plants. *Front. Plant Sci.* **2018**, *9*, 140. <https://doi.org/10.3389/fpls.2018.00140>.
6. Sweet, D.; Tirado, S.; Springer, N.; Hirsch, C.; Hirsch, C. Opportunities and challenges in phenotyping row crops using drone-based RGB imaging. *Plant Phenome J.* **2022**, *5*, e20044. <https://doi.org/10.1002/ppj2.20044>.
7. Kamilaris, A.; Prenafeta-Boldú, F.X. Deep learning in agriculture: A survey. *Comput. Electron. Agric.* **2018**, *147*, 70–90. <https://doi.org/10.1016/j.compag.2018.02.016>.
8. Wang, N.; Fu, S.; Rao, Q.; Zhang, G.; Ding, M. Insect-YOLO: A new method of crop insect detection. *Comput. Electron. Agric.* **2025**, *232*, 110085. <https://doi.org/10.1016/j.compag.2025.110085>.
9. Su, W.-H.; Zhang, J.; Yang, C.; Page, R.; Szinyei, T.; Hirsch, C.D.; Steffenson, B.J. Automatic evaluation of wheat resistance to Fusarium head blight using dual Mask-RCNN deep learning frameworks in computer vision. *Remote Sens.* **2021**, *13*, 26. <https://doi.org/10.3390/rs13010026>.
10. Barbedo, J.G.A. Data fusion in agriculture: Resolving ambiguities and closing data gaps. *Sensors* **2022**, *22*, 2285. <https://doi.org/10.3390/s22062285>.
11. Auat Cheein, F.A.; Carelli, R. Agricultural robotics: Unmanned robotic service units in agricultural tasks. *IEEE Ind. Electron. Mag.* **2013**, *7*, 48–58. <https://doi.org/10.1109/MIE.2013.2252957>.
12. Lu, D.; Wang, Y. MAR-YOLOv9: A multi-dataset object detection method for agricultural fields based on YOLOv9. *PLoS ONE* **2024**, *19*, e0307643. <https://doi.org/10.1371/journal.pone.0307643>.
13. Attari, I.; Awasthi, L.K.; Sharma, T.P.; Rathee, P. A review of deep learning techniques used in agriculture. *Ecol. Inform.* **2023**, *77*, 102217. <https://doi.org/10.1016/j.ecoinf.2023.102217>.
14. Zhang, C.; Li, T.; Li, J. Detection of impurity rate of machine-picked cotton based on improved Canny operator. *Electronics* **2022**, *11*, 974. <https://doi.org/10.3390/electronics11070974>.
15. Bora, D.J.; Gupta, A.K.; Khan, F.A. Comparing the performance of LAB* and HSV color spaces with respect to color image segmentation. *arXiv* **2015**, arXiv:1506.01472. <https://doi.org/10.48550/arXiv.1506.01472>.
16. Ronneberger, O.; Fischer, P.; Brox, T. U-Net: Convolutional networks for biomedical image segmentation. *arXiv* **2015**, arXiv:1505.04597. <https://doi.org/10.48550/arXiv.1505.04597>.
17. He, K.; Gkioxari, G.; Dollár, P.; Girshick, R. Mask R-CNN. *arXiv* **2017**, arXiv:1703.06870. <https://doi.org/10.48550/arXiv.1703.06870>.
18. Li, Y.; Guo, R.; Li, R.; Ji, R.; Wu, M.; Chen, D.; Han, C.; Han, R.; Liu, Y.; Ruan, Y.; et al. An improved U-net and attention mechanism-based model for sugar beet and weed segmentation. *Front. Plant Sci.* **2024**, *15*, 1449514. <https://doi.org/10.3389/fpls.2024.1449514>.
19. Knott, M.; Odion, D.; Sontakke, S.; Karwa, A.; Defraeye, T. Weakly supervised image segmentation for defect-based grading of fresh produce. *arXiv* **2024**, arXiv:2411.16219. <https://doi.org/10.48550/arXiv.2411.16219>.
20. Hu, J.; Shen, L.; Sun, G. Squeeze-and-excitation networks. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Salt Lake City, UT, USA, 18–23 June 2018; IEEE: Piscataway, NJ, USA, 2018; pp. 7132–7141. <https://doi.org/10.1109/CVPR.2018.00745>.
21. Devi, P.S.; Rajan, A.S. An inquiry of image processing in agriculture to perceive plant disease using machine learning. *Multimed. Tools Appl.* **2024**, *83*, 80631–80640. <https://doi.org/10.1007/s11042-023-18052-4>.
22. Benmouna, B.; Pourdarbani, R.; Sabzi, S.; Fernandez-Beltran, R.; García-Mateos, G.; Molina-Martínez, J.M. Attention mechanisms in convolutional neural networks for nitrogen treatment detection in tomato leaves using hyperspectral images. *Electronics* **2023**, *12*, 2706. <https://doi.org/10.3390/electronics12122706>.

23. Ren, S.; He, K.; Girshick, R.; Sun, J. Faster R-CNN: Towards real-time object detection with region proposal networks. *IEEE Trans. Pattern Anal. Mach. Intell.* **2017**, *39*, 1137–1149. <https://doi.org/10.1109/TPAMI.2016.2577031>.
24. Hou, B. Theoretical analysis of the network structure of two mainstream object detection methods: YOLO and Fast R-CNN. *Appl. Comput. Eng.* **2023**, *17*, 213–225. <https://doi.org/10.54254/2755-2721/17/20230943>.
25. Jiang, K.; Xie, T.; Yan, R.; Wen, X.; Li, D.; Jiang, H.; Jiang, N.; Feng, L.; Duan, X.; Wang, J. An Attention mechanism-improved YOLOv7 algorithm for hemp duck count estimation. *Agriculture* **2022**, *12*, 1659. <https://doi.org/10.3390/agriculture12101659>.
26. Liu, J.; Abbas, I.; Noor, R.S. Development of deep learning-based variable rate agrochemical spraying system for targeted weeds control in strawberry crop. *Agronomy* **2021**, *11*, 1480. <https://doi.org/10.3390/agronomy11081480>.
27. Lv, R.; Hu, J.; Zhang, T.; Chen, X.; Liu, W. Crop-free-ridge navigation line recognition based on the lightweight structure improvement of YOLOv8. *Agriculture* **2025**, *15*, 942. <https://doi.org/10.3390/agriculture15090942>.
28. Huang, X.-Y.; Pan, S.-H.; Sun, Z.-Y.; Ye, W.-T.; Aheto, J.H. Evaluating quality of tomato during storage using fusion information of computer vision and electronic nose. *J. Food Process Eng.* **2018**, *41*, e12832. <https://doi.org/10.1111/jfpe.12832>.
29. Chen, J.; Zhang, M.; Xu, B.; Sun, J.; Mujumdar, A.S. Artificial intelligence assisted technologies for controlling the drying of fruits and vegetables using physical fields: A review. *Trends Food Sci. Technol.* **2020**, *105*, 251–260. <https://doi.org/10.1016/j.tifs.2020.08.015>.
30. Pei, H.; Sun, Y.; Huang, H.; Zhang, W.; Sheng, J.; Zhang, Z. Weed detection in maize fields by UAV images based on crop row preprocessing and improved YOLOv4. *Agriculture* **2022**, *12*, 975. <https://doi.org/10.3390/agriculture12070975>.
31. Zheng, K.; Zhao, X.; Han, C.; He, Y.; Zhai, C.; Zhao, C. Design and experiment of an automatic row-oriented spraying system based on machine vision for early-stage maize crops. *Agriculture* **2023**, *13*, 691. <https://doi.org/10.3390/agriculture13030691>.
32. Zhang, T.; Zhou, J.; Liu, W.; Yue, R.; Yao, M.; Shi, J.; Hu, J. Seedling-YOLO: High-efficiency target detection algorithm for field broccoli seedling transplanting quality based on YOLOv7-tiny. *Agronomy* **2024**, *14*, 931. <https://doi.org/10.3390/agronomy14050931>.
33. Zhang, W.; Cao, H.; Zhang, W.; Hanan, J.S.; Ge, D.; Cao, J.; Xia, J.; Xuan, S.; Liang, W.; Zhang, L.; et al. An aboveground biomass partitioning coefficient model for rapeseed (*Brassica napus* L.). *Field Crops Res.* **2020**, *259*, 107966. <https://doi.org/10.1016/j.fcr.2020.107966>.
34. Jiang, L.; Li, C.; Fu, L. Apple tree architectural trait phenotyping with organ-level instance segmentation from point cloud. *Comput. Electron. Agric.* **2025**, *229*, 109708. <https://doi.org/10.1016/j.compag.2024.109708>.
35. Yuan, Q.; Wang, P.; Luo, W.; Zhou, Y.; Chen, H.; Meng, Z. Simultaneous localization and mapping system for agricultural yield estimation based on improved VINS-RGBD: A case study of a strawberry field. *Agriculture* **2024**, *14*, 784. <https://doi.org/10.3390/agriculture14050784>.
36. Guo, H.; Bao, W.; Qu, K.; Ma, X.; Cao, M. Multispectral and hyperspectral image fusion based on regularized coupled non-negative block-term tensor decomposition. *Remote Sens.* **2022**, *14*, 5306. <https://doi.org/10.3390/rs14215306>.
37. Ye, Z.; Zhai, X.; She, T.; Liu, X.; Hong, Y.; Wang, L.; Zhang, L.; Wang, Q. Winter wheat yield prediction based on the ASTGNN model coupled with multi-source data. *Agronomy* **2024**, *14*, 2262. <https://doi.org/10.3390/agronomy14102262>.
38. NY/T 1301–2007; Technical Regulations for Regional Trials of Crop Varieties—Wheat. China Agriculture Press: Beijing, China, 2007. Available online: <https://std.samr.gov.cn/hb/search/stdHBDetailed?id=B28FFD881CB4C494E05397BE0A0AB84E> (accessed on 13 March 2026).
39. Pun Magar, L.; Sandifer, J.; Khatri, D.; Poudel, S.; KC, S.; Gyawali, B.; Gebremedhin, M.; Chiluwal, A. Plant height measurement using UAV-based aerial RGB and LiDAR images in soybean. *Front. Plant Sci.* **2025**, *16*, 1488760. <https://doi.org/10.3389/fpls.2025.1488760>.
40. Jimenez-Berni, J.A.; Deery, D.M.; Rozas-Larraondo, P.; Condon, A.G.; Rebetzke, G.J.; James, R.A.; Bovill, W.D.; Furbank, R.T.; Sirault, X.R.R. High-throughput determination of plant height, ground cover, and above-ground biomass in wheat with LiDAR. *Front. Plant Sci.* **2018**, *9*, 237. <https://doi.org/10.3389/fpls.2018.00237>.
41. Kim, W.-S.; Lee, D.-H.; Kim, Y.-J.; Kim, Y.-S.; Kim, T.; Park, S.-U.; Kim, S.-S.; Hong, D.-H. Crop height measurement system based on 3D image and tilt sensor fusion. *Agronomy* **2020**, *10*, 1670. <https://doi.org/10.3390/agronomy10111670>.
42. Li, Y.; Li, C.; Cheng, Q.; Chen, L.; Li, Z.; Zhai, W.; Mao, B.; Chen, Z. Precision estimation of winter wheat crop height and above-ground biomass using unmanned aerial vehicle imagery and oblique photography point cloud data. *Front. Plant Sci.* **2024**, *15*, 1437350. <https://doi.org/10.3389/fpls.2024.1437350>.
43. GB/T 5262–2008; General Provisions for Test Conditions and Methods for Agricultural Machinery. Standards Press of China: Beijing, China, 2008. Available online: <https://std.samr.gov.cn/gb/search/gbDetailed?id=71F772D76D82D3A7E05397BE0A0AB82A> (accessed on 13 March 2026).

44. Velumani, K.; Lopez-Lozano, R.; Madec, S.; Guo, W.; Gillet, J.; Comar, A.; Baret, F. Estimates of maize plant density from UAV RGB images using Faster R-CNN detection model: Impact of the spatial resolution. *Plant Phenomics* **2021**, 2021, 9824843. <https://doi.org/10.34133/2021/9824843>.
45. Li, Y.; Zhang, X.; Chen, D. CSNet: Dilated convolutional neural networks for understanding highly congested scenes. *arXiv* **2018**, arXiv:1802.10062. <https://doi.org/10.48550/arXiv.1802.10062>.
46. Peng, J.; Rezaei, E.E.; Zhu, W.; Wang, D.; Li, H.; Yang, B.; Sun, Z. Plant density estimation using UAV imagery and deep learning. *Remote Sens.* **2022**, *14*, 5923. <https://doi.org/10.3390/rs14235923>.
47. Lu, N.; Zhou, J.; Han, Z.; Li, D.; Cao, Q.; Yao, X.; Tian, Y.; Zhu, Y.; Cao, W.; Cheng, T. Improved estimation of aboveground biomass in wheat from RGB imagery and point cloud data acquired with a low-cost unmanned aerial vehicle system. *Plant Methods* **2019**, *15*, 17. <https://doi.org/10.1186/s13007-019-0402-3>.
48. Jin, X.; Liu, S.; Baret, F.; Hemmerlé, M.; Comar, A. Estimates of plant density of wheat crops at emergence from very low altitude UAV imagery. *Remote Sens. Environ.* **2017**, *198*, 105–114. <https://doi.org/10.1016/j.rse.2017.06.007>.
49. Xu, X.; Geng, Q.; Gao, F.; Xiong, D.; Qiao, H.; Ma, X. Segmentation and counting of wheat spike grains based on deep learning and textural feature. *Plant Methods* **2023**, *19*, 77. <https://doi.org/10.1186/s13007-023-01062-6>.
50. Shi, L.; Sun, J.; Dang, Y.; Zhang, S.; Sun, X.; Xi, L.; Wang, J. YOLOv5s-T: A lightweight small object detection method for wheat spikelet counting. *Agriculture* **2023**, *13*, 872. <https://doi.org/10.3390/agriculture13040872>.
51. Qiu, R.; He, Y.; Zhang, M. Automatic detection and counting of wheat spikelet using semi-automatic labeling and deep learning. *Front. Plant Sci.* **2022**, *13*, 872555. <https://doi.org/10.3389/fpls.2022.872555>.
52. Sanaeifar, A.; Guindo, M.L.; Bakhshipour, A.; Fazayeli, H.; Li, X.; Yang, C. Advancing precision agriculture: The potential of deep learning for cereal plant head detection. *Comput. Electron. Agric.* **2023**, *209*, 107875. <https://doi.org/10.1016/j.compag.2023.107875>.
53. NY/T 4379–2023; General Technical Specification for Agricultural Remote Sensing Survey—Crop Lodging Monitoring. China Agriculture Press: Beijing, China, 2023. Available online: <https://std.samr.gov.cn/hb/search/stdHBDetailed?id=FF13AEF5E762F62CE05397BE0A0A5929> (accessed on 13 March 2026).
54. Wen, J.; Yin, Y.; Zhang, Y.; Pan, Z.; Fan, Y. Detection of wheat lodging by binocular cameras during harvesting operation. *Agriculture* **2023**, *13*, 120. <https://doi.org/10.3390/agriculture13010120>.
55. Kang, G.; Wang, J.; Zeng, F.; Cai, Y.; Kang, G.; Yue, X. Lightweight detection system with global attention network (GloAN) for rice lodging. *Plants* **2023**, *12*, 1595. <https://doi.org/10.3390/plants12081595>.
56. Wilke, N.; Siegmund, B.; Klingbeil, L.; Burkart, A.; Rascher, U. Quantifying lodging percentage and lodging severity using a UAV-based canopy height model combined with an objective threshold approach. *Remote Sens.* **2019**, *11*, 515. <https://doi.org/10.3390/rs11050515>.
57. Xie, B.; Wang, J.; Jiang, H.; Zhao, S.; Liu, J.; Jin, Y.; Li, Y. Multi-feature detection of in-field grain lodging for adaptive low-loss control of combine harvesters. *Comput. Electron. Agric.* **2023**, *208*, 107772. <https://doi.org/10.1016/j.compag.2023.107772>.
58. Li, Z.; Feng, X.; Li, J.; Wang, D.; Hong, W.; Qin, J.; Wang, A.; Ma, H.; Yao, Q. Time series field estimation of rice canopy height using an unmanned aerial vehicle-based RGB/multispectral platform. *Agronomy* **2024**, *14*, 883. <https://doi.org/10.3390/agronomy14050883>.
59. Perna, C.; Pagliai, A.; Sarri, D.; Lisci, R.; Vieri, M. Can a Light Detection and Ranging (LiDAR) and Multispectral Sensor Discriminate Canopy Structure Changes Due to Pruning in Olive Growing? A Field Experimentation. *Sensors* **2024**, *24*, 7894. <https://doi.org/10.3390/s24247894>.
60. Shu, M.; Li, Q.; Ghafoor, A.; Zhu, J.; Li, B.; Ma, Y. Using plant height and canopy coverage to estimate maize aboveground biomass with UAV digital images. *Eur. J. Agron.* **2023**, *151*, 126957. <https://doi.org/10.1016/j.eja.2023.126957>.
61. Xie, J.; Zhou, Z.; Zhang, H.; Zhang, L.; Li, M. Combining canopy coverage and plant height from UAV-based RGB images to estimate spraying volume on potato. *Sustainability* **2022**, *14*, 6473. <https://doi.org/10.3390/su14116473>.
62. Valluvan, A.B.; Raj, R.; Pingale, R.; Jagarlapudi, A. Canopy height estimation using drone-based RGB images. *Smart Agric. Technol.* **2023**, *4*, 100145. <https://doi.org/10.1016/j.atech.2022.100145>.
63. Joshi, A.; Pradhan, B.; Gite, S.; Chakraborty, S. Remote-sensing data and deep-learning techniques in crop mapping and yield prediction: A systematic review. *Remote Sens.* **2023**, *15*, 2014. <https://doi.org/10.3390/rs15082014>.
64. Shi, J.; Bai, Y.; Diao, Z.; Zhou, J.; Yao, X.; Zhang, B. Row detection-based navigation and guidance for agricultural robots and autonomous vehicles in row-crop fields: Methods and applications. *Agronomy* **2023**, *13*, 1780. <https://doi.org/10.3390/agronomy13071780>.

65. Zhang, S.; Ma, Q.; Cheng, S.; An, D.; Yang, Z.; Ma, B.; Yang, Y. Crop row detection in the middle and late periods of maize under sheltering based on solid-state LiDAR. *Agriculture* **2022**, *12*, 2011. <https://doi.org/10.3390/agriculture12122011>.
66. Wang, Q.; Meng, Z.-J.; Wen, C.-K.; Qin, W.-C.; Wang, F.; Zhang, A.-Q.; Zhao, C.-J.; Yin, Y.-X. Grain combine harvester header profiling control system development and testing. *Comput. Electron. Agric.* **2024**, *223*, 109082. <https://doi.org/10.1016/j.compag.2024.109082>.
67. Zhou, J.; Cui, M.; Wu, Y.; Gao, Y.; Tang, Y.; Chen, Z.; Hou, L.; Tian, H. Maize (*Zea mays* L.) stem target region extraction and stem diameter measurement based on an internal gradient algorithm in field conditions. *Agronomy* **2023**, *13*, 1185. <https://doi.org/10.3390/agronomy13051185>.
68. Lehnert, C.; McCool, C.; Perez, T. In-field peduncle detection of sweet peppers for robotic harvesting: A comparative study. *arXiv* **2017**, arXiv:1709.10275. <https://doi.org/10.48550/arXiv.1709.10275>.
69. Wu, F.; Wang, J.; Zhou, Y.; Song, X.; Ju, C.; Sun, C.; Liu, T. Estimation of winter wheat tiller number based on optimization of gradient vegetation characteristics. *Remote Sens.* **2022**, *14*, 1338. <https://doi.org/10.3390/rs14061338>.
70. Ma, J.; Li, M.; Fan, W.; Liu, J. State-of-the-art techniques for fruit maturity detection. *Agronomy* **2024**, *14*, 2783. <https://doi.org/10.3390/agronomy14122783>.
71. Mahmood, A.; Singh, S.K.; Tiwari, A.K. Pre-trained deep learning-based classification of jujube fruits according to their maturity level. *Neural Comput. Appl.* **2022**, *34*, 13925–13935. <https://doi.org/10.1007/s00521-022-07213-5>.
72. Zhao, M.; Cang, H.; Chen, H.; Zhang, C.; Yan, T.; Zhang, Y.; Gao, P.; Xu, W. Determination of quality and maturity of processing tomatoes using near-infrared hyperspectral imaging with interpretable machine learning methods. *LWT* **2023**, *183*, 114861. <https://doi.org/10.1016/j.lwt.2023.114861>.
73. Tzuan, G.T.H.; Hashim, F.H.; Raj, T.; Baseri Huddin, A.; Sajab, M.S. Oil palm fruits ripeness classification based on Raman spectra characteristics. *Plants* **2022**, *11*, 1936. <https://doi.org/10.3390/plants11151936>.
74. DB6111/T 175–2021; Smart Agriculture—Technical Specification for Three-Dimensional Digital Data Acquisition and Modeling of Fruits. China Standards Press: Beijing, China, 2021. Available online: <https://std.samr.gov.cn/db/search/stdDBDetailed?id=EB32634DC009404BE05397BE0A0AEE0D> (accessed on 13 March 2026).
75. Beldek, C.; Dunn, A.; Cunningham, J.; Sariyildiz, E.; Phung, S.L.; Alici, G. Multi-vision-based picking point localisation of target fruit for harvesting robots. *arXiv* **2025**, arXiv:2502.12406. <https://doi.org/10.48550/arXiv.2502.12406>.
76. Kang, H.; Wang, X.; Chen, C. Real-Time Fruit Recognition and Grasping Estimation for Robotic Apple Harvesting. *Sensors* **2020**, *20*, 5670. <https://doi.org/10.3390/s20195670>.
77. Liu, T.; Wang, X.; Hu, K.; Zhou, H.; Kang, H.; Chen, C. FF3D: A rapid and accurate 3D fruit detector for robotic harvesting. *Sensors* **2024**, *24*, 3858. <https://doi.org/10.3390/s24123858>.
78. Lowe, A.; Harrison, N.; French, A.P. Hyperspectral image analysis techniques for the detection and classification of the early onset of plant disease and stress. *Plant Methods* **2017**, *13*, 80. <https://doi.org/10.1186/s13007-017-0233-z>.
79. Yang, N.; Yu, J.; Wang, A.; Tang, J.; Zhang, R.; Xie, L.; Shu, F.; Oppong Paul, K. A rapid rice blast detection and identification method based on crop disease spores' diffraction fingerprint texture. *J. Sci. Food Agric.* **2020**, *100*, 3608–3621. <https://doi.org/10.1002/jsfa.10383>.
80. Zhu, W.; Sun, J.; Wang, S.; Shen, J.; Yang, K.; Zhou, X. Identifying field crop diseases using transformer-embedded convolutional neural network. *Agriculture* **2022**, *12*, 1083. <https://doi.org/10.3390/agriculture12081083>.
81. Zhang, Y.; Guo, J.; Bian, F.; Li, Z.; Guo, C.; Zheng, J.; Zhang, X. Crop disease spore detection method based on Au@Ag NRS. *Agriculture* **2025**, *15*, 2076. <https://doi.org/10.3390/agriculture15192076>.
82. Yang, N.; Yuan, M.; Wang, P.; Zhang, R.; Sun, J.; Mao, H. Tea diseases detection based on fast infrared thermal image processing technology. *J. Sci. Food Agric.* **2019**, *99*, e9564. <https://doi.org/10.1002/jsfa.9564>.
83. Zhao, S.; Peng, Y.; Liu, J.; Wu, S. Tomato leaf disease diagnosis based on improved convolution neural network by attention module. *Agriculture* **2021**, *11*, 651. <https://doi.org/10.3390/agriculture11070651>.
84. Nagasubramanian, K.; Jones, S.; Singh, A.K.; Sarkar, S.; Singh, A.; Ganapathysubramanian, B. Plant disease identification using explainable 3D deep learning on hyperspectral images. *Plant Methods* **2019**, *15*, 98. <https://doi.org/10.1186/s13007-019-0479-8>.
85. Chiou, K.D.; Chen, Y.X.; Chen, P.S.; Jou, Y.-T.; Tsai, S.-H.; Chang, C.-Y. Application of deep learning for fruit defect recognition in *Psidium guajava* L. *Sci. Rep.* **2025**, *15*, 6145. <https://doi.org/10.1038/s41598-025-88936-y>.
86. Pang, Q.; Huang, W.; Fan, S.; Zhou, Q.; Wang, Z.; Tian, X. Detection of early bruises on apples using hyperspectral imaging combined with YOLOv3 deep learning algorithm. *J. Food Process Eng.* **2022**, *45*, e13952. <https://doi.org/10.1111/jfpe.13952>.
87. Chu, P.; Li, Z.; Zhang, K.; Lammers, K.; Lu, R. High-precision fruit localization using active laser–camera scanning: Robust laser line extraction for 2D–3D transformation. *arXiv* **2023**, arXiv:2311.08674. <https://doi.org/10.48550/arXiv.2311.08674>.

88. Yue, J.; Zhou, C.; Feng, H.; Yang, Y.; Zhang, N. Novel applications of optical sensors and machine learning in agricultural monitoring. *Agriculture* **2023**, *13*, 1970. <https://doi.org/10.3390/agriculture13101970>.
89. Wang, W.; den Brinker, A.C. Modified RGB cameras for infrared remote-PPG. *IEEE Trans. Biomed. Eng.* **2020**, *67*, 2893–2904. <https://doi.org/10.1109/TBME.2020.2973313>.
90. Linhares, J.M.M.; Monteiro, J.A.R.; Bailão, A.; Cardeira, L.; Kondo, T.; Nakauchi, S.; Picollo, M.; Cucci, C.; Casini, A.; Stefani, L.; et al. How good are RGB cameras retrieving colors of natural scenes and paintings?—A study based on hyperspectral imaging. *Sensors* **2020**, *20*, 6242. <https://doi.org/10.3390/s20216242>.
91. Zhang, J.; Yang, C.; Song, H.; Hoffmann, W.C.; Zhang, D.; Zhang, G. Evaluation of an airborne remote sensing platform consisting of two consumer-grade cameras for crop identification. *Remote Sens.* **2016**, *8*, 257. <https://doi.org/10.3390/rs8030257>.
92. Yan, B.; Li, X. RGB-D camera and fractal-geometry-based maximum diameter estimation method of apples for robot intelligent selective graded harvesting. *Fractal Fract.* **2024**, *8*, 649. <https://doi.org/10.3390/fractalfract8110649>.
93. Fan, X.; Zhang, Y.; Zhou, S.; Ren, M.; Wang, Y.; Chai, X. Recognition and position method of tomato picking robot based on improved YOLOv8s and RGB-D information fusion. *Trans. Chin. Soc. Agric. Eng.* **2025**, *41*, 106–116. <https://doi.org/10.11975/j.issn.1002-6819.202503181>.
94. Zhang, S.; Qiu, B.; Xue, X.; Sun, T.; Peng, B. Parameters optimization of crop protection UAS based on the first industry standard of China. *Int. J. Agric. Biol. Eng.* **2020**, *13*, 29–35. <https://doi.org/10.25165/j.ijabe.20201303.5439>.
95. Liu, W.; Hu, J.; Liu, J.; Yue, R.; Zhang, T.; Yao, M.; Li, J. Method for the navigation line recognition of the ridge without crops via machine vision. *Int. J. Agric. Biol. Eng.* **2024**, *17*, 230–239. <https://doi.org/10.25165/j.ijabe.20241702.7480>.
96. Niu, Y.; Han, W.; Zhang, H.; Zhang, L.; Chen, H. Estimating maize plant height using a crop surface model constructed from UAV RGB images. *Biosyst. Eng.* **2024**, *241*, 56–67. <https://doi.org/10.1016/j.biosystemseng.2024.04.003>.
97. Zhang, L.; Wang, A.; Zhang, H.; Zhu, Q.; Zhang, H.; Sun, W. Estimating leaf chlorophyll content of winter wheat from UAV multispectral images using machine learning algorithms under different species, growth stages, and nitrogen stress conditions. *Agriculture* **2024**, *14*, 1064. <https://doi.org/10.3390/agriculture14071064>.
98. Wei, L.; Yang, H.; Niu, Y.; Zhang, Y.; Xu, L.; Chai, X. Wheat biomass, yield, and straw-grain ratio estimation from multi-temporal UAV-based RGB and multispectral images. *Biosyst. Eng.* **2023**, *234*, 187–205. <https://doi.org/10.1016/j.biosystemseng.2023.08.002>.
99. Zheng, Z.; Yuan, J.; Yao, W.; Kwan, P.; Yao, H.; Liu, Q.; Guo, L. Fusion of UAV-acquired visible images and multispectral data by applying machine-learning methods in crop classification. *Agronomy* **2024**, *14*, 2670. <https://doi.org/10.3390/agronomy14112670>.
100. Mafuratidze, P.; Chibarabada, T.P.; Shekede, M.D.; Masocha, M. A new four-stage approach based on normalized vegetation indices for detecting and mapping sugarcane hail damage using multispectral remotely sensed data. *Geocarto Int.* **2023**, *38*, 2245788. <https://doi.org/10.1080/10106049.2023.2245788>.
101. Lee, J.; Park, Y.; Kim, H.; Yoon, Y.-Z.; Ko, W.; Bae, K.; Lee, J.-Y.; Choo, H.; Roh, Y.-G. Compact meta-spectral image sensor for mobile applications. *Nanophotonics* **2022**, *11*, 2563–2569. <https://doi.org/10.1515/nanoph-2021-0706>.
102. Zhang, J.; Rivard, B.; Rogge, D.M. The successive projection algorithm (SPA), an algorithm with a spatial constraint for the automatic search of endmembers in hyperspectral data. *Sensors* **2008**, *8*, 1321–1342. <https://doi.org/10.3390/s8021321>.
103. Aheto, J.H.; Huang, X.; Tian, X.; Ren, Y.; Bonah, E.; Alenyorege, E.A.; Lv, R.; Dai, C. Combination of spectra and image information of hyperspectral imaging data for fast prediction of lipid oxidation attributes in pork meat. *J. Food Process Eng.* **2019**, *42*, e13225. <https://doi.org/10.1111/jfpe.13225>.
104. Elsayed, S.; El-Hendawy, S.; Elsherbiny, O.; Okasha, A.M.; Elmetwalli, A.H.; Elwakeel, A.E.; Memon, M.S.; Ibrahim, M.E.M.; Ibrahim, H.H. Estimating chlorophyll content, production, and quality of sugar beet under various nitrogen levels using machine learning models and novel spectral indices. *Agronomy* **2023**, *13*, 2743. <https://doi.org/10.3390/agronomy13112743>.
105. Cong, S.; Sun, J.; Shi, L.; Dai, C.; Wu, X.; Zhang, B.; Yao, K.; Ge, X. Nondestructive assessment of refrigerated quality attributes of kiwifruit via hyperspectral imaging coupled with a point-to-interval prediction model. *Food Chem.* **2026**, *499*, 147323. <https://doi.org/10.1016/j.foodchem.2025.147323>.
106. Zhang, Y.; Sun, J.; Li, J.; Wu, X.; Dai, C. Quantitative analysis of cadmium content in tomato leaves based on hyperspectral image and feature selection. *Appl. Eng. Agric.* **2018**, *34*, 789–798. <https://doi.org/10.13031/aea.12679>.
107. Xu, M.; Sun, J.; Yao, K.; Wu, X.; Shen, J.; Cao, Y.; Zhou, X. Nondestructive detection of total soluble solids in grapes using VMD-RC and hyperspectral imaging. *J. Food Sci.* **2022**, *87*, 326–338. <https://doi.org/10.1111/1750-3841.16004>.
108. Tian, Y.; Sun, J.; Zhou, X.; Yao, K.; Tang, N. Detection of soluble solid content in apples based on hyperspectral technology combined with deep learning algorithm. *J. Food Process. Preserv.* **2022**, *46*, e16414. <https://doi.org/10.1111/jfpp.16414>.

109. Zhao, J.; Li, H.; Chen, C.; Pang, Y.; Zhu, X. Detection of water content in lettuce canopies based on hyperspectral imaging technology under outdoor conditions. *Agriculture* **2022**, *12*, 1796. <https://doi.org/10.3390/agriculture12111796>.
110. Zhu, W.; Feng, Z.; Dai, S.; Zhang, P.; Wei, X. Using UAV multispectral remote sensing with appropriate spatial resolution and machine learning to monitor wheat scab. *Agriculture* **2022**, *12*, 1785. <https://doi.org/10.3390/agriculture12111785>.
111. Cong, C.; Cao, G.; Li, Y.; Liu, D.; Ma, B.; Zhang, J.; Li, L.; Hu, J. Research on monitoring methods for the appropriate rice harvest period based on multispectral remote sensing. *Discret. Dyn. Nat. Soc.* **2022**, *2022*, 1519667. <https://doi.org/10.1155/2022/1519667>.
112. Sun, J.; Lu, X.; Mao, H.; Wu, X.; Gao, H. Quantitative determination of rice moisture based on hyperspectral imaging technology and BCC-LS-SVR algorithm. *J. Food Process Eng.* **2017**, *40*, e12446. <https://doi.org/10.1111/jfpe.12446>.
113. Wu, P.; Sun, S.; Wei, W.; Yuan, X.; Zhou, R. Research on calibration methods of long-wave infrared camera and visible camera. *J. Sens.* **2022**, *2022*, 8667606. <https://doi.org/10.1155/2022/8667606>.
114. Kang, R.; Ma, T.; Tsuchikawa, S.; Inagaki, T.; Chen, J.; Zhao, J.; Li, D.; Cui, G. Non-destructive near-infrared moisture detection of dried goji (*Lycium barbarum* L.) berry. *Horticulturae* **2024**, *10*, 302. <https://doi.org/10.3390/horticulturae10030302>.
115. Guo, Z.; Chen, X.; Zhang, Y.; Sun, C.; Jayan, H.; Majeed, U.; Watson, N.J.; Zou, X. Dynamic nondestructive detection models of apple quality in critical harvest period based on near-infrared spectroscopy and intelligent algorithms. *Foods* **2024**, *13*, 1698. <https://doi.org/10.3390/foods13111698>.
116. Kathirvelan, J.; Vijayaraghavan, R. An infrared-based sensor system for the detection of ethylene for the discrimination of fruit ripening. *Infrared Phys. Technol.* **2017**, *85*, 403–409. <https://doi.org/10.1016/j.infrared.2017.07.022>.
117. Lee, Y.H.; Khalil-Hani, M.; Bakhteri, R.; Nambiar, V.P. A real-time near infrared image acquisition system based on image quality assessment. *J. Real-Time Image Process.* **2017**, *13*, 103–120. <https://doi.org/10.1007/s11554-016-0586-y>.
118. Shen, T.; Zou, X.; Shi, J.; Li, Z.; Huang, X.; Xu, Y.; Chen, W. Determination geographical origin and flavonoids content of goji berry using near-infrared spectroscopy and chemometrics. *Food Anal. Methods* **2016**, *9*, 68–79. <https://doi.org/10.1007/s12161-015-0175-x>.
119. Zhou, X.; Sun, J.; Zhang, Y.; Tian, Y.; Yao, K.; Xu, M. Visualization of heavy metal cadmium in lettuce leaves based on wavelet support vector machine regression model and visible-near infrared hyperspectral imaging. *J. Food Process Eng.* **2021**, *44*, e13897. <https://doi.org/10.1111/jfpe.13897>.
120. Mahmood ur Rehman, M.; Liu, J.; Nijabat, A.; Faheem, M.; Wang, W.; Zhao, S. Leveraging convolutional neural networks for disease detection in vegetables: A comprehensive review. *Agronomy* **2024**, *14*, 2231. <https://doi.org/10.3390/agronomy14102231>.
121. Liu, X.; Zhang, L.; Zhai, X.; Li, L.; Zhou, Q.; Chen, X.; Li, X. Polarization LiDAR: Principles and applications. *Photonics* **2023**, *10*, 1118. <https://doi.org/10.3390/photonics10101118>.
122. Tan, H.; Wang, P.; Yan, X.; Xin, Q.; Mu, G.; Lv, Z. A highly accurate detection platform for potato seedling canopy in intelligent agriculture based on phased array LiDAR technology. *Agriculture* **2024**, *14*, 1369. <https://doi.org/10.3390/agriculture14081369>.
123. Xu, W.; Yang, W.; Wu, J.; Chen, P.; Lan, Y.; Zhang, L. Canopy laser interception compensation mechanism—UAV LiDAR precise monitoring method for cotton height. *Agronomy* **2023**, *13*, 2584. <https://doi.org/10.3390/agronomy13102584>.
124. Yuan, W.; Choi, D.; Bolkas, D. GNSS-IMU-assisted colored ICP for UAV-LiDAR point cloud registration of peach trees. *Comput. Electron. Agric.* **2022**, *197*, 106966. <https://doi.org/10.1016/j.compag.2022.106966>.
125. Qi, C.R.; Su, H.; Mo, K.; Guibas, L.J. PointNet: Deep learning on point sets for 3D classification and segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI, USA, 21–26 July 2017; IEEE: Piscataway, NJ, USA, 2017; pp. 652–660. <https://doi.org/10.1109/CVPR.2017.16>.
126. Murcia, H.F.; Tilaguy, S.; Ouazaa, S. Development of a low-cost system for 3D orchard mapping integrating UGV and LiDAR. *Plants* **2021**, *10*, 2804. <https://doi.org/10.3390/plants10122804>.
127. Reji, J.; Nidamanuri, R.R. Deep learning-based fusion of LiDAR point cloud and multispectral imagery for crop classification sensitive to nitrogen level. In *Proceedings of the International Conference on Machine Intelligence for GeoAnalytics and Remote Sensing (MIGARS)*; IEEE: Piscataway, NJ, USA, 2023; pp. 1–4. <https://doi.org/10.1109/MIGARS57353.2023.10064497>.
128. Zhao, Y.; Zhang, X.; Sun, J.; Yu, T.; Cai, Z.; Zhang, Z.; Mao, H. Low-cost lettuce height measurement based on depth vision and lightweight instance segmentation model. *Agriculture* **2024**, *14*, 1596. <https://doi.org/10.3390/agriculture14091596>.
129. Jones, H.G. Application of thermal imaging and infrared sensing in plant physiology and ecophysiology. In *Advances in Botanical Research*; Academic Press: London, UK, 2004; Volume 41, pp. 107–163. [https://doi.org/10.1016/S0065-2296\(04\)41003-9](https://doi.org/10.1016/S0065-2296(04)41003-9).
130. Wei, Z.; Zhang, F.; Chang, S.; Liu, Y.; Wu, H.; Feng, Z. MmWave radar and vision fusion for object detection in autonomous driving: A review. *arXiv* **2021**, arXiv:2108.03004. <https://doi.org/10.48550/arXiv.2108.03004>.

131. Xu, S.; Xu, X.; Zhu, Q.; Meng, Y.; Yang, G.; Feng, H.; Yang, M.; Zhu, Q.; Xue, H.; Wang, B. Monitoring leaf nitrogen content in rice based on information fusion of multi-sensor imagery from UAV. *Precis. Agric.* **2023**, *24*, 2327–2349. <https://doi.org/10.1007/s11119-023-10042-8>.
132. Zhang, L.; Zhang, B.; Zhang, H.; Yang, W.; Hu, X.; Cai, J.; Wu, C.; Wang, X. Multi-source feature fusion network for LAI estimation from UAV multispectral imagery. *Agronomy* **2025**, *15*, 988. <https://doi.org/10.3390/agronomy15040988>.
133. Akkem, Y.; Biswas, S.K.; Varanasi, A. Smart farming using artificial intelligence: A review. *Eng. Appl. Artif. Intell.* **2023**, *120*, 105899. <https://doi.org/10.1016/j.engappai.2023.105899>.
134. Nitin; Gupta, S.B.; Yadav, R.; Bovand, F.; Tyagi, P.K. Developing precision agriculture using data augmentation framework for automatic identification of castor insect pests. *Front. Plant Sci.* **2023**, *14*, 1101943. <https://doi.org/10.3389/fpls.2023.1101943>.
135. Liu, M.; Chen, J.; Han, X. Research on Retinex algorithm combining with attention mechanism for image enhancement. *Electronics* **2022**, *11*, 3695. <https://doi.org/10.3390/electronics11223695>.
136. Zheng, J.; Xu, C.; Zhang, W.; Yang, X. Single image dehazing using global illumination compensation. *Sensors* **2022**, *22*, 4169. <https://doi.org/10.3390/s22114169>.
137. Avola, G.; Matese, A.; Riggi, E. An overview of the special issue on “Precision Agriculture Using Hyperspectral Images”. *Remote Sens.* **2023**, *15*, 1917. <https://doi.org/10.3390/rs15071917>.
138. Yang, G.; Yang, H.; Yu, S.; Wang, J.; Nie, Z. A multi-scale dehazing network with dark channel priors. *Sensors* **2023**, *23*, 5980. <https://doi.org/10.3390/s23135980>.
139. Liu, Z.; Xiao, G.; Liu, H.; Wei, H. Multi-sensor measurement and data fusion. *IEEE Instrum. Meas. Mag.* **2022**, *25*, 28–36. <https://doi.org/10.1109/MIM.2022.9693406>.
140. Guan, Z.; Li, H.; Chen, X.; Mu, S.; Jiang, T.; Zhang, M.; Wu, C. Development of impurity-detection system for tracked rice combine harvester based on DEM and Mask R-CNN. *Sensors* **2022**, *22*, 9550. <https://doi.org/10.3390/s22239550>.
141. Chen, W.-M.; Tsai, H.-H.; Ling, J.F. Parallel computation of dominance scores for multidimensional datasets on GPUs. *IEEE Trans. Parallel Distrib. Syst.* **2024**, *35*, 919–931. <https://doi.org/10.1109/TPDS.2024.3382119>.
142. Shen, L.; Su, J.; He, R.; Song, L.; Huang, R.; Fang, Y.; Song, Y.; Su, B. Real-time tracking and counting of grape clusters in the field based on channel pruning with YOLOv5s. *Comput. Electron. Agric.* **2023**, *206*, 107662. <https://doi.org/10.1016/j.compag.2023.107662>.
143. Jani, M.; Fayyad, J.; Al-Younes, Y.; Najjaran, H. Model compression methods for YOLOv5: A review. *arXiv* **2023**, arXiv:2307.11904. <https://doi.org/10.48550/arXiv.2307.11904>.
144. Lu, J.; Hu, J.; Zhao, G.; Mei, F.; Zhang, C. An in-field automatic wheat disease diagnosis system. *arXiv* **2017**, arXiv:1709.10275. <https://doi.org/10.48550/arXiv.1710.08299>.
145. Feng, Y.; Shen, A.; Hu, J.; Liang, Y.; Wang, S.; Du, J. Enhancing few-shot learning with integrated data and GAN model approaches. *arXiv* **2024**, arXiv:2411.16567. <https://doi.org/10.48550/arXiv.2411.16567>.
146. Ma, Z.; Wang, X.; Chen, X.; Hu, B.; Li, J. Advances in crop row detection for agricultural robots: Methods, performance indicators, and scene adaptability. *Agriculture* **2025**, *15*, 2151. <https://doi.org/10.3390/agriculture15202151>.
147. Jin, Y.; Liu, J.; Xu, Z.; Yuan, S.; Li, P.; Wang, J. Development status and trend of agricultural robot technology. *Int. J. Agric. Biol. Eng.* **2021**, *14*, 1–19. <https://doi.org/10.25165/j.ijabe.20211404.6821>.

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